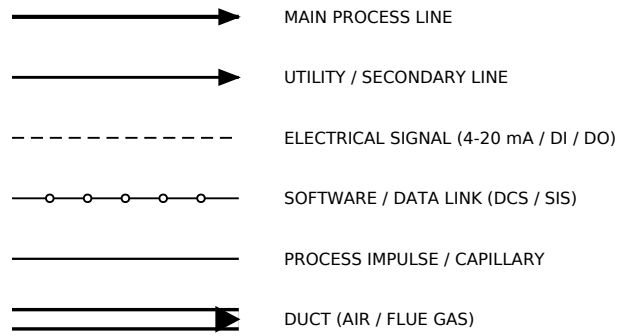
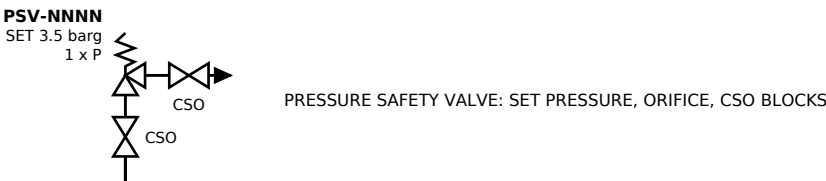
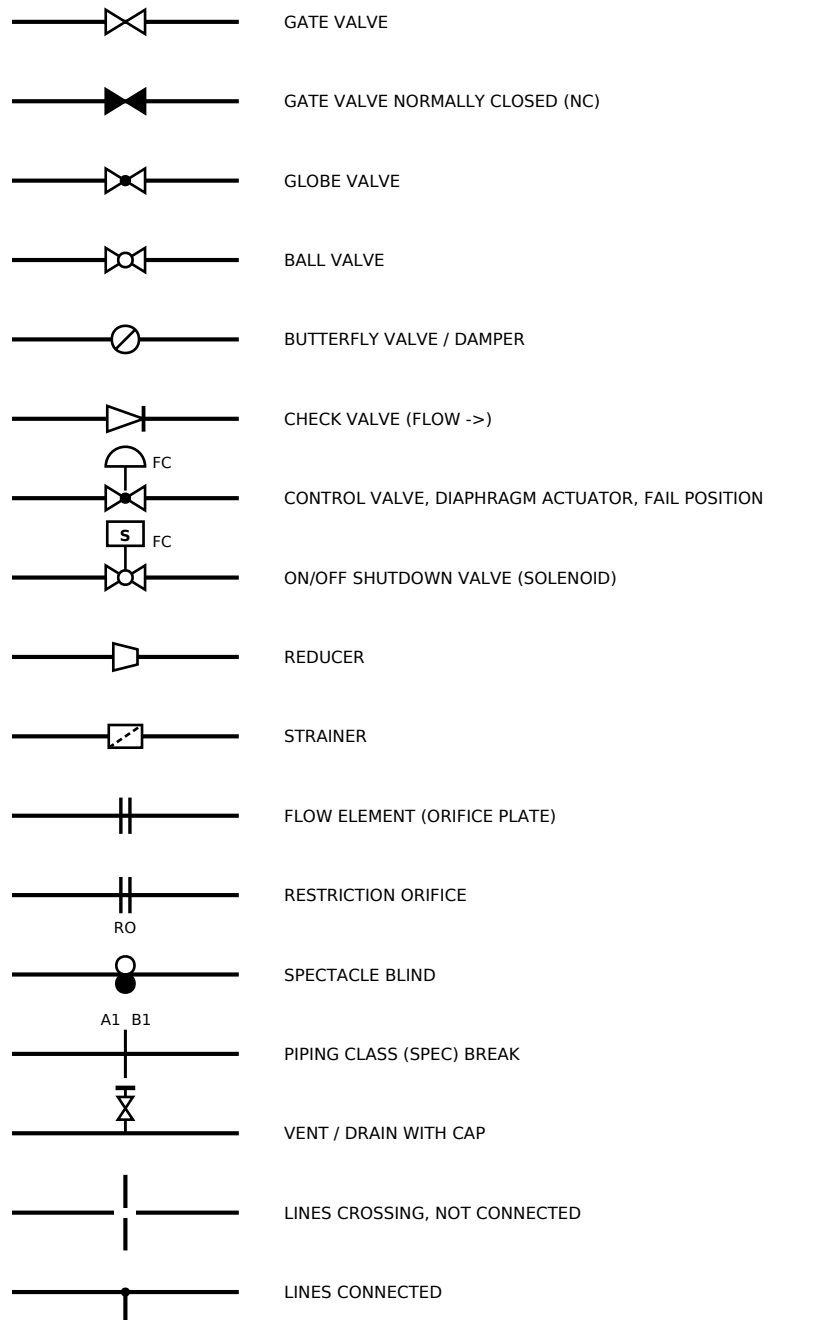


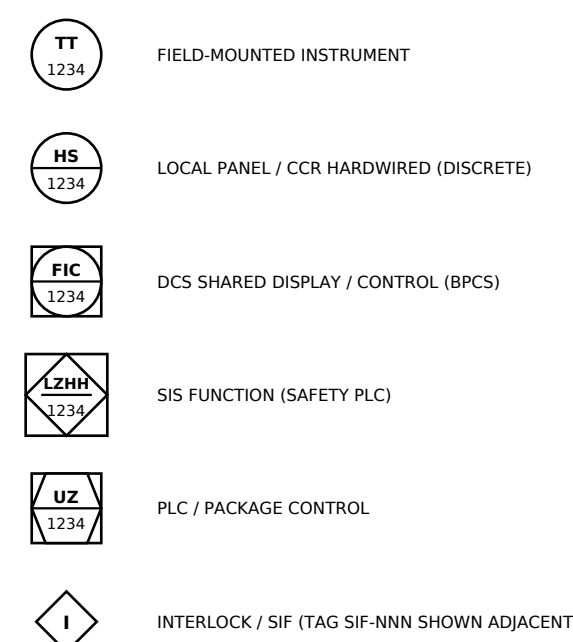
LINE SYMBOLS



VALVES & PIPING COMPONENTS



INSTRUMENT SYMBOLS (ISA-5.1)



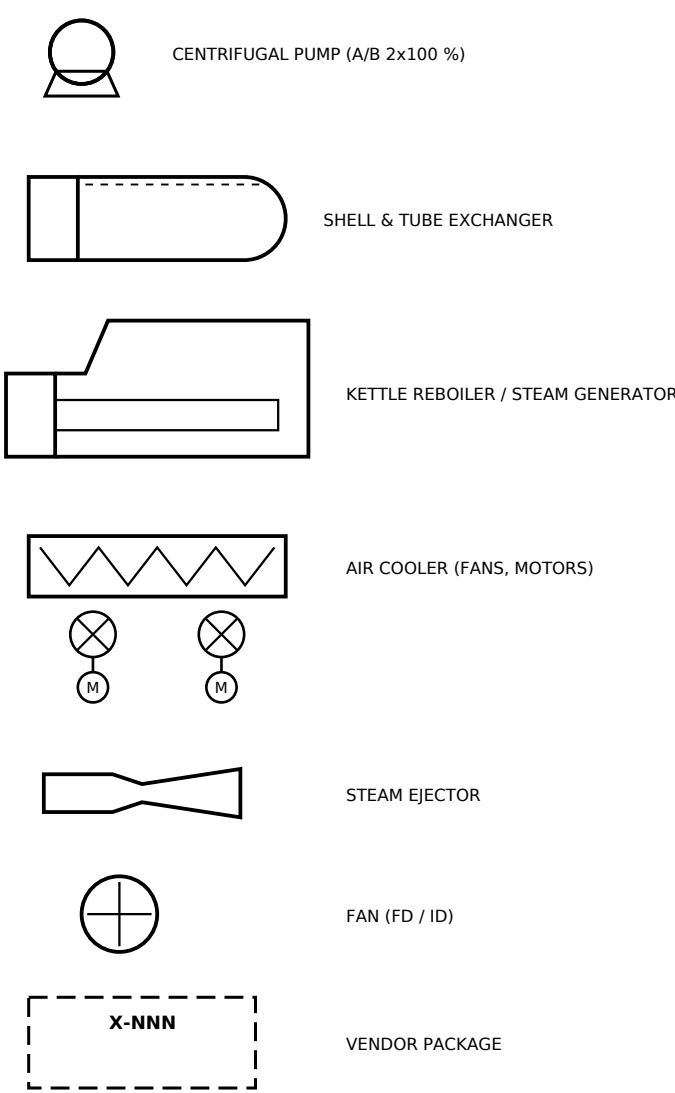
INSTRUMENT IDENTIFICATION LETTERS

A ANALYSIS (O<sub>2</sub>, pH) B BURNER / FLAME  
F FLOW H HAND  
L LEVEL P PRESSURE  
T TEMPERATURE X/Z ON-OFF / POSITION  
PD / TD DIFFERENTIAL FF FLOW RATIO  
Succeeding: E element, G glass, I indicate, C control,  
T transmit, V valve, Y relay/compute, S switch, Z (2x4) SIS,  
AH/AL/AH/ALL alarms; A/B/C suffix = redundant SIS

ABBREVIATIONS

FC / FO / FL FAIL CLOSED / OPEN / LAST  
CSO / CSC CAR-SEALED OPEN / CLOSED  
NC / NO NORMALLY CLOSED / OPEN  
NMF NORMALLY NO FLOW  
TPA / MPA / BPA TOP / MID / BOTTOM PUMPAROUND  
SS STRIPPING STEAM  
DBB DOUBLE BLOCK & BLEED  
ROSOV / EIV REMOTE OPERATED SHUT-OFF VALVE  
COT / COT COIL OUTLET / INLET TEMP.  
RO RESTRICTION ORIFICE  
FA FLAME ARRESTOR  
OSBL OUTSIDE BATTERY LIMITS  
BMS BURNER MANAGEMENT SYSTEM  
2oo3 TWO-OUT-OF-THREE VOTING

EQUIPMENT SYMBOLS



LINE NUMBER

12"-P-100-002-B1-H

NPS (in) - FLUID - AREA - SEQUENCE - PIPING CLASS - INSULATION  
AREA: 100 CDU, 200 VDU, 900 UNIT UTILITIES

FLUID CODES

P Process hydrocarbon  
PG Process gas / vapour  
SW Sour water  
WW Wash water / brine  
LS LP steam  
HS HP steam  
CD Condensate  
BFW Boiler feed water

CWS Cooling water supply  
CWR Cooling water return  
FG Fuel gas  
FL Flare  
BD Blowdown / drain  
IA Instrument air  
N Nitrogen  
CH Chemical

INSULATION CODES

H HEAT CONSERVATION  
P PERSONNEL PROTECTION  
ST STEAM TRACED + INSULATED  
C COLD  
N NONE

PIPING CLASSES (SUMMARY, SEE CFU-000-PI-SPC-001)

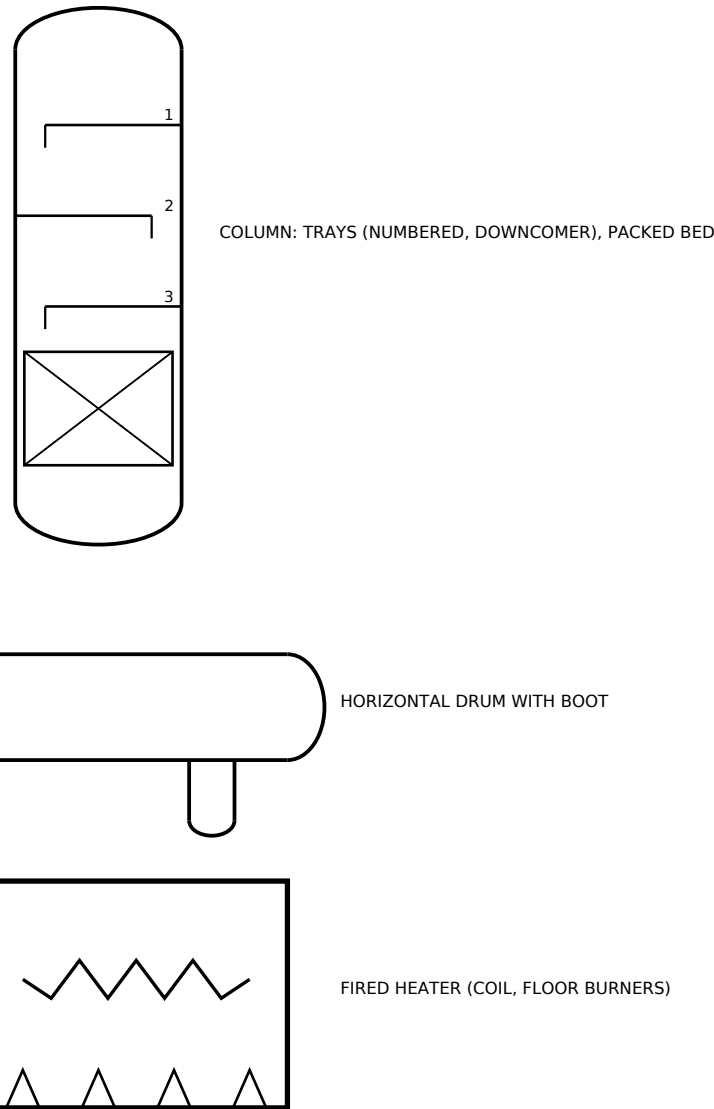
A1 150# Carbon steel ASTM A106 Gr.B  
A2 150# Carbon steel, HIC-resistant (NACE MR0103)  
B1 300# Carbon steel ASTM A106 Gr.B  
B2 300# 5Cr-1/2Mo ASTM A335 P5  
B3 300# 9Cr-1Mo ASTM A335 P9  
C1 600# Carbon steel, killed ASTM A106 Gr.B  
S1 150# Carbon steel ASTM A106 Gr.B  
S2 600# 1.25Cr-1/2Mo ASTM A335 P11  
U1 150# Carbon steel (cement-lined for CW >= 12")  
U2 150# S5304 (IA) / galvanised CS (N2)  
V1 150# 5Cr-1/2Mo ASTM A335 P5

Hydrocarbon <= 230 C, general  
Sour water, OH vapour/condensate, wash water/brine  
Hydrocarbon <= 260 C, pump discharges  
Sulfidic hydrocarbon 260-400 C (incl crude, AR, BPA, HvGO, VR)  
Heater outlets / transfer lines (H-101, H-201) and > 400 C  
LPG / stabiliser, BFW > 300#  
LP / MP steam, condensate  
HP steam > 400 C  
Cooling water, utility water, LP BFW  
Instrument air, nitrogen  
Vacuum vapour lines (C-201 OH); transfer line > 400 C -> B3

DRAWING INDEX

CFU-000-PR-PID-000 LEGEND, SYMBOLS & ABBREVIATIONS  
CFU-100-PR-PID-001 CRUDE CHARGE PUMPS & COLD PREHEAT TRAIN  
CFU-100-PR-PID-002 ELECTROSTATIC DESALTERS D-101A/B & BOOSTER PUMPS  
CFU-100-PR-PID-003 HOT PREHEAT TRAIN E-106 TO E-111 & E-113  
CFU-100-PR-PID-004 ATMOSPHERIC HEATER H-101 - PROCESS COILS  
CFU-100-PR-PID-005 ATMOSPHERIC HEATER H-101 - FIRING & AIR SYSTEM  
CFU-100-PR-PID-006 ATMOSPHERIC COLUMN C-101 - FLASH ZONE & BOTTOMS  
CFU-100-PR-PID-007 ATMOSPHERIC COLUMN C-101 - UPPER SECTION & PUMPAROUNDS  
CFU-100-PR-PID-008 SIDE STRIPPERS C-102/103/104 & PRODUCT RUNDOWN  
CFU-100-PR-PID-009 C-101 OVERHEAD SYSTEM & REFLUX DRUM D-102  
CFU-100-PR-PID-010 NAPHTHA STABILISER C-105  
CFU-100-PR-PID-011 NAPHTHA SPLITTER C-106  
CFU-200-PR-PID-012 VACUUM HEATER H-201  
CFU-200-PR-PID-013 VACUUM COLUMN C-201  
CFU-200-PR-PID-014 VACUUM PUMPAROUNDS & PRODUCTS  
CFU-200-PR-PID-015 VACUUM EJECTOR SYSTEM & HOTWELL D-201  
CFU-900-PR-PID-016 FUEL GAS, FLARE KO & UTILITY HEADERS

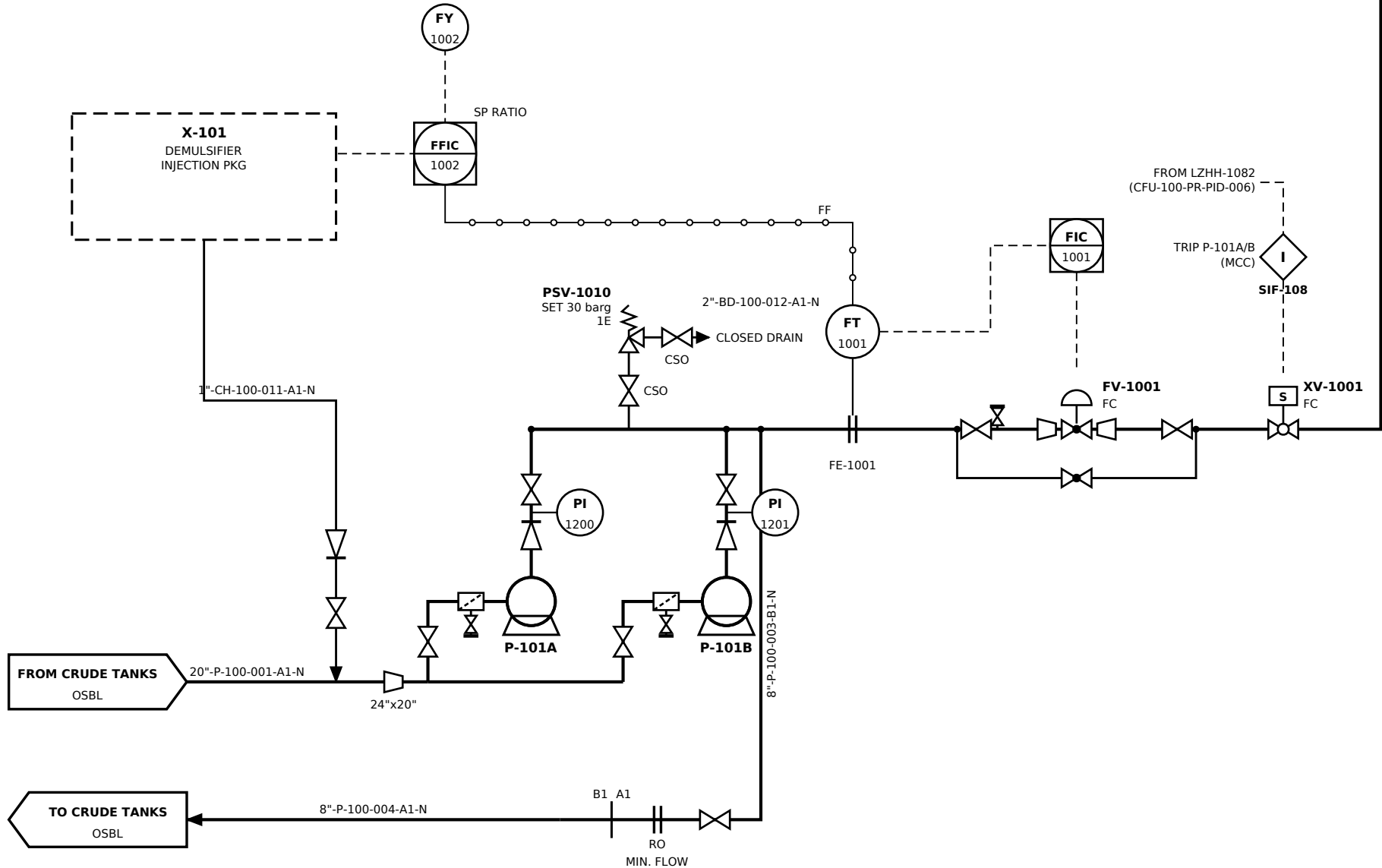
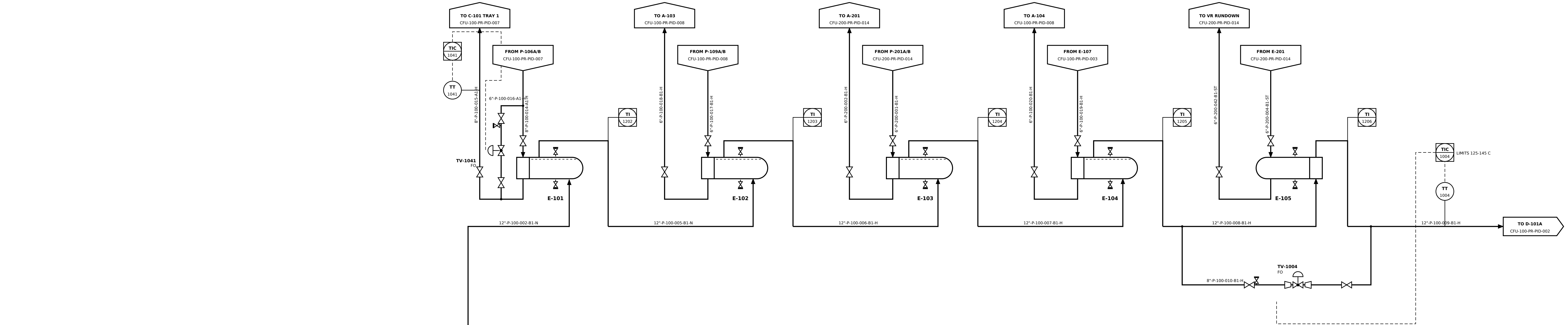
EQUIPMENT DATA BOX



NOTES  
1. Legend applies to all P&IDs CFU-xxx-PR-PID-001 to -016.

| REV                                                                         | DATE       | DESCRIPTION              |            |
|-----------------------------------------------------------------------------|------------|--------------------------|------------|
| A                                                                           | 2026-10-02 | Issued for review (FEED) |            |
| GENERIC REFINERY - GULF COAST<br>100 KBPSD CRUDE & VACUUM DISTILLATION UNIT |            |                          |            |
| PIPING & INSTRUMENT DIAGRAM<br>LEGEND, SYMBOLS & ABBREVIATIONS              |            |                          |            |
| DISCIPLINE                                                                  | STAGE      | SCALE                    | SHEET      |
| PROCESS                                                                     | FEED       | NTS                      | 1 OF 17    |
| DRAWN                                                                       | CHECKED    | APPROVED                 | DATE       |
| CLAUDE CODE                                                                 | -          | -                        | 2026-10-02 |
| DRAWING NUMBER                                                              |            |                          | REV        |
| CFU-000-PR-PID-000                                                          |            |                          | A          |

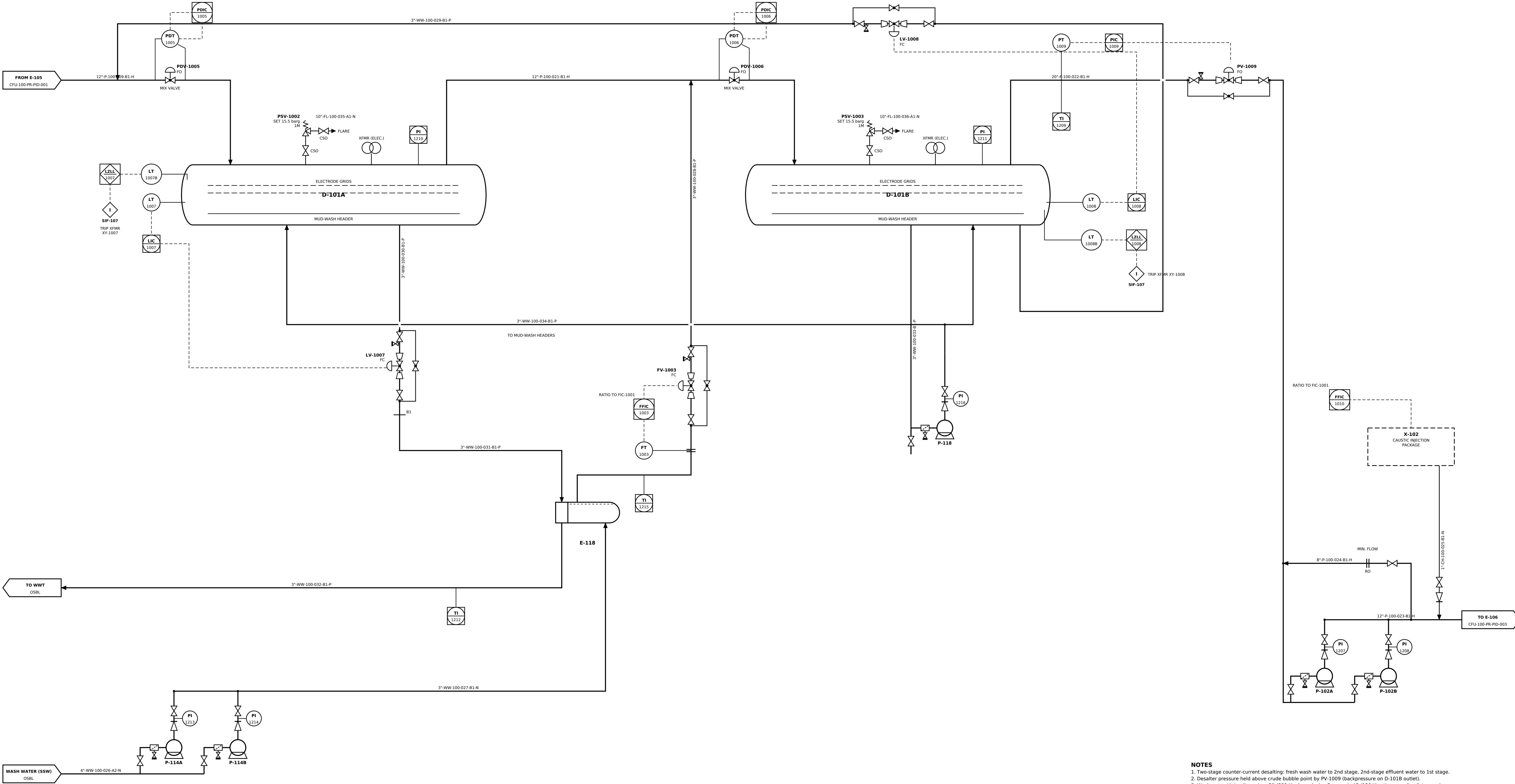
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|-----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|
| <b>P-101A/B</b><br>Crude charge<br>RATED 730 m3/h @ 261 m, dP 22 bar<br>MOTOR 710 kW, API 682, Flan 11/22<br>DES. T 60 C; MAT'L S-6 (CS/12Cr) | <b>E-101</b><br>Crude / Top pumparound<br>359 m2 total; 15 x 1P shells<br>DESIGN 30 / 20 barg / 185 C; CA 3 mm<br>DUTY 8.09 MW<br>SHELL: Crude / TUBE: Top pumparound<br>MAT'L CS shell / CS tubes | <b>E-102</b><br>Crude / Kerosene product<br>438 m2 total; 15 x 1P shells<br>DESIGN 30 / 20 barg / 240 C; CA 3 mm<br>DUTY 6.55 MW<br>SHELL: Crude / TUBE: Kerosene product<br>MAT'L CS shell / CS tubes | <b>E-103</b><br>Crude / LVGO pumparound/product<br>804 m2 total; 25 x 1P shells<br>DESIGN 30 / 20 barg / 255 C; CA 3 mm<br>DUTY 11.45 MW<br>SHELL: Crude / TUBE: LVGO pumparound/product<br>MAT'L CS shell / CS tubes | <b>E-104</b><br>Crude / Diesel product<br>378 m2 total; 15 x 1P shells<br>DESIGN 30 / 20 barg / 235 C; CA 3 mm<br>DUTY 4.44 MW<br>SHELL: Crude / TUBE: Diesel product<br>MAT'L CS shell / CS tubes | <b>E-105</b><br>Crude / Vacuum residue<br>431 m2 total; 15 x 1P shells<br>DESIGN 30 / 20 barg / 285 C; CA 3 mm<br>DUTY 5.84 MW<br>SHELL: Vacuum residue / TUBE: Crude<br>MAT'L CS shell / 5Cr-1/2Mo tubes | <b>X-101</b><br>Demulsifier injection package<br>Tank 2 m3 + 2 x 100 % metering pumps<br>MOTOR 0.75 kW<br>MAT'L SS316 |
|-----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|



- NOTES**
- Crude-side design pressure 30 barg (= PSV-1010 set) for P-101 discharge system to D-101A inlet.
  - Exchanger hot-side connections shown at channel / shell nozzles; TEMA type & shells per data sheets.
  - TV-1041 / TV-1004 bypass control valves have block valves only (exchanger provides bypass path).
  - Demulsifier dosed at P-101 suction, ratio to FIC-1001 (FFIC-1002) - X-101 vendor package.
  - Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
  - All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
  - Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

| REV                                                                         | DATE       | DESCRIPTION              |            |
|-----------------------------------------------------------------------------|------------|--------------------------|------------|
| A                                                                           | 2026-10-02 | Issued for review (FEED) |            |
| GENERIC REFINERY - GULF COAST<br>100 KBPSD CRUDE & VACUUM DISTILLATION UNIT |            |                          |            |
| PIPING & INSTRUMENT DIAGRAM<br>CRUDE CHARGE PUMPS & COLD PREHEAT TRAIN      |            |                          |            |
| DISCIPLINE                                                                  | STAGE      | SCALE                    | SHEET      |
| PROCESS                                                                     | FEED       | NTS                      | 2 OF 17    |
| DRAWN                                                                       | CHECKED    | APPROVED                 | DATE       |
| CLAUDE CODE                                                                 | -          | -                        | 2026-10-02 |
| DRAWING NUMBER                                                              |            |                          | REV        |
| CFU-100-PR-PID-001                                                          |            |                          | A          |

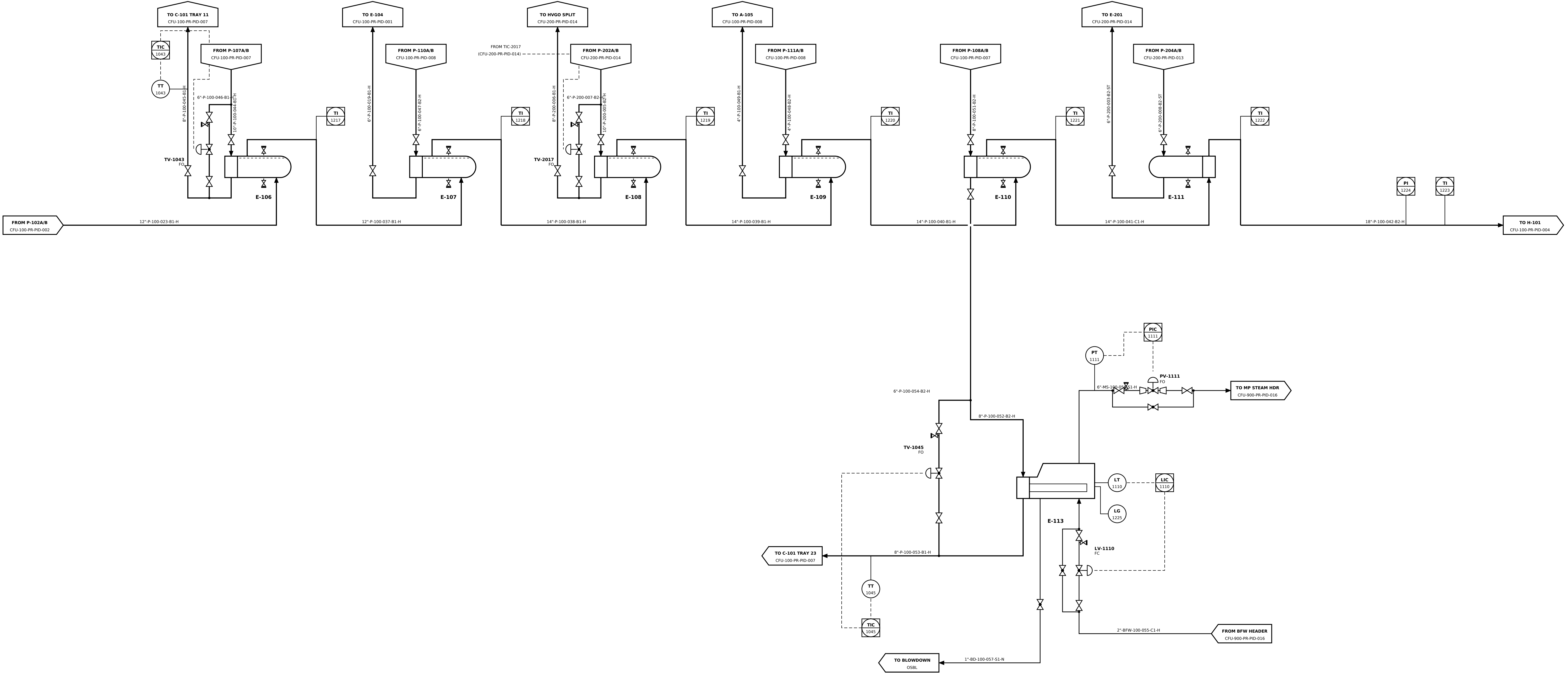
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|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>D-101A</b><br>Electrostatic desalter, 1st stage<br>ID 3.8 m x 30.0 m TTT, ACDC dual-polarity grids<br>DESIGN 17.0 barg / 165 C, CA 3 mm<br>MAT'L CS | <b>D-101B</b><br>Electrostatic desalter, 2nd stage<br>ID 3.8 m x 30.0 m TTT, ACDC dual-polarity grids<br>DESIGN 17.0 barg / 165 C, CA 3 mm<br>MAT'L CS | <b>E-118</b><br>Desalter wash water / brine<br>84 m2, 1 shell(s)<br>DESIGN 20 / 20 barg / 160 C, CA 3 mm<br>DUTY 1.62 MW<br>SHELL: Wash water / TUBE: Brine<br>MAT'L CS shell / duplex 2205 tubes | <b>P-102A/B</b><br>Desalted crude booster<br>RATED 805 m3/h @ 263 m, dP 20 bar<br>MOTOR 719 kW, API 682 Plan 11/52<br>DES. T 165 C, MAT'L S-6 (CS/12Cr) | <b>P-114A/B</b><br>Desalter wash water<br>RATED 37 m3/h @ 82 m, dP 8 bar<br>MOTOR 18.3 kW, API 682 Plan 11/52<br>DES. T 80 C, MAT'L S-6 (CS/12Cr) | <b>P-118</b><br>Desalter mud-wash / recycle<br>RATED 19 m3/h @ 64 m, dP 6 bar<br>MOTOR 7.5 kW, API 682 Plan 11/52<br>DES. T 150 C, MAT'L S-6 (CS/12Cr) | <b>X-102</b><br>Caustic injection package (desalted crude)<br>Tank 2 m3 + 2 x 100 % metering pumps<br>MOTOR 0.75 kW<br>MAT'L SS316 |
|--------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|



- NOTES**
- Two-stage counter-current desalting; fresh wash water to 2nd stage, 2nd-stage effluent water to 1st stage.
  - Desalter pressure held above crude bubble point by PV-1009 (backpressure on D-101B outlet).
  - Wash water / brine lines class B1 (300#) where design P exceeds A2 (150#) rating - see line list remarks.
  - Transformer trip on low-low interface (SIF-107) by Electrical (XY-1007 / XY-1008 to feeder CB).
  - Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
  - All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
  - Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

| REV                                                                             | DATE       | DESCRIPTION              |            |
|---------------------------------------------------------------------------------|------------|--------------------------|------------|
| A                                                                               | 2026-10-02 | Issued for review (FEED) |            |
| GENERIC REFINERY - GULF COAST<br>100 KBPSD CRUDE & VACUUM DISTILLATION UNIT     |            |                          |            |
| PIPING & INSTRUMENT DIAGRAM<br>ELECTROSTATIC DESALTERS D-101A/B & BOOSTER PUMPS |            |                          |            |
| DISCIPLINE                                                                      | STAGE      | SCALE                    | SHEET      |
| PROCESS                                                                         | FEED       | NTS                      | 3 OF 17    |
| DRAWN                                                                           | CHECKED    | APPROVED                 | DATE       |
| CLAUDE CODE                                                                     | -          | -                        | 2026-10-02 |
| DRAWING NUMBER                                                                  |            |                          | REV        |
| CFU-100-PR-PID-002                                                              |            |                          | A          |

|                                                                                                                                                                                                                               |                                                                                                                                                                                                           |                                                                                                                                                                                                                                |                                                                                                                                                                                                      |                                                                                                                                                                                                                                    |                                                                                                                                                                                                             |                                                                                                                                                                                               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>E-106</b><br>Crude / Middle (kero) pumparound<br>1001 m2 total; 25 x 1P shells<br>DESIGN 30 / 20 barg / 270 C; CA 3 mm<br>DUTY 5.83 MW<br>SHELL: Crude / TUBE: Middle (kero) pumparound<br>MATL CS shell / 5Cr-1/2Mo tubes | <b>E-107</b><br>Crude / Diesel product<br>324 m2 total; 15 x 1P shells<br>DESIGN 30 / 20 barg / 325 C; CA 6 mm<br>DUTY 12.85 MW<br>SHELL: Crude / TUBE: Diesel product<br>MATL CS shell / 5Cr-1/2Mo tubes | <b>E-108</b><br>Crude / HVGO pumparound/product<br>2196 m2 total; 25 x 2P shells<br>DESIGN 30 / 20 barg / 345 C; CA 6 mm<br>DUTY 21.14 MW<br>SHELL: Crude / TUBE: HVGO pumparound/product<br>MATL 1.25Cr shell / 9Cr-1Mo tubes | <b>E-109</b><br>Crude / AGO product<br>263 m2 total; 15 x 1P shells<br>DESIGN 30 / 20 barg / 365 C; CA 6 mm<br>DUTY 3.11 MW<br>SHELL: Crude / TUBE: AGO product<br>MATL 1.25Cr shell / 9Cr-1Mo tubes | <b>E-110</b><br>Crude / Bottom (diesel) pumparound<br>886 m2 total; 15 x 2P shells<br>DESIGN 30 / 20 barg / 345 C; CA 6 mm<br>DUTY 7.46 MW<br>SHELL: Crude / TUBE: Bottom (diesel) pumparound<br>MATL 1.25Cr shell / 9Cr-1Mo tubes | <b>E-111</b><br>Crude / Vacuum residue<br>1094 m2 total; 15 x 2P shells<br>DESIGN 30 / 20 barg / 395 C; CA 6 mm<br>DUTY 7.58 MW<br>SHELL: Vacuum residue / TUBE: Crude<br>MATL 1.25Cr shell / 9Cr-1Mo tubes | <b>E-113</b><br>BPA / MP steam generator (kettle)<br>223 m2; 1 shell(s)<br>DESIGN 13 / 20 barg / 290 C; CA 3 mm<br>DUTY 4.91 MW<br>SHELL: BFW / MP steam / TUBE: Bottom pumparound<br>MATL CS |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



- NOTES**
- Hot train crude-side design pressure per P-102 shut-off (see line list); exchanger rating to be confirmed.
  - E-113 kettle MP steam generator: 3-element level control simplified to LIC-1110 at FEED.
  - TV-2017 controlled from TIC-2017 (HVGO pumparound return temperature) on CFU-200-PR-PID-014.
  - Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
  - All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
  - Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

| REV                                                                         | DATE       | DESCRIPTION              |             |
|-----------------------------------------------------------------------------|------------|--------------------------|-------------|
| A                                                                           | 2026-10-02 | Issued for review (FEED) |             |
| GENERIC REFINERY - GULF COAST<br>100 KBPSD CRUDE & VACUUM DISTILLATION UNIT |            |                          |             |
| PIPING & INSTRUMENT DIAGRAM<br>HOT PREHEAT TRAIN E-106 TO E-111 & E-113     |            |                          |             |
| DISCIPLINE                                                                  |            | STAGE                    | SHEET       |
| PROCESS                                                                     |            | FEED                     | NTS 4 OF 17 |
| DRAWN                                                                       | CHECKED    | APPROVED                 | DATE        |
| CLAUDE CODE                                                                 | -          | -                        | 2026-10-02  |
| DRAWING NUMBER                                                              |            |                          | REV         |
| CFU-100-PR-PID-003                                                          |            |                          | A           |

**H-101**  
Atmospheric crude charge heater  
60.7 MW abs. / 67.4 MW fired; 2-cell cabin, 8 passes, 144 rad. tubes 6" x 18.3 m  
DESIGN 38.5 barg / 545 C; CA 3 mm  
DUTY 60.68 MW  
MAT'L Tubes A335 P9 (9Cr-1Mo); convection P5 / CS studded

LP STEAM HEADER  
CFU-900-PR-PID-016

FROM E-111 (CIT)  
CFU-100-PR-PID-003

BFW HEADER  
CFU-900-PR-PID-016  
1" BFW-100-072-016-H

TV-1024  
FC

TIC  
1024

350 C  
TT  
1024

PI  
1227

STRIPPING STEAM HDR  
CFU-100-PR-PID-006

TT  
1020A-C

2003  
TSW  
1025

SIF-106

TO C-101 FLASH ZONE  
CFU-100-PR-PID-006

TT  
1020

TIC  
1026

SP CASCADE TO PIC-1021  
(CFU-100-PR-PID-005); FF FROM FIC-1001

24"x28"

TDIC  
1019

BIAS TO FIC-1011-1018  
(TOTAL FLOW HELD)

18" P-100-042-B2-H

2003 PER PASS  
LOW-LOW PASS FLOW

FT  
1011

FIC  
1011

FV-1011  
FO

FT  
1011A-C

FZLL  
1011

6" P-100-058-B2-H

FT  
1012

FIC  
1012

FV-1012  
FO

FT  
1012A-C

FZLL  
1012

6" P-100-059-B2-H

FT  
1013

FIC  
1013

FV-1013  
FO

FT  
1013A-C

FZLL  
1013

6" P-100-060-B2-H

FT  
1014

FIC  
1014

FV-1014  
FO

FT  
1014A-C

FZLL  
1014

6" P-100-061-B2-H

FT  
1015

FIC  
1015

FV-1015  
FO

FT  
1015A-C

FZLL  
1015

6" P-100-062-B2-H

FT  
1016

FIC  
1016

FV-1016  
FO

FT  
1016A-C

FZLL  
1016

6" P-100-063-B2-H

FT  
1017

FIC  
1017

FV-1017  
FO

FT  
1017A-C

FZLL  
1017

6" P-100-064-B2-H

FT  
1018

FIC  
1018

FV-1018  
FO

FT  
1018A-C

FZLL  
1018

6" P-100-065-B2-H

H-101

16 BURNERS (FLOOR-FIRED, 2 CELLS) - SEE CFU-100-PR-PID-005

- NOTES**
- Passes shown schematically (convection + radiant in series per pass); 8 identical passes.
  - Tube-skin thermocouples: 4 per pass (TI-1101..1132) - shown on heater vendor P&ID, alarmed in DCS.
  - Pass flow SIS transmitters FT-10xA/B/C (2003 per pass) independent from BPCS FT-10x (SIF-101).
  - Decoking (steam-air) connections and spool pieces at inlet / outlet manifolds by heater vendor.
  - Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
  - All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
  - Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

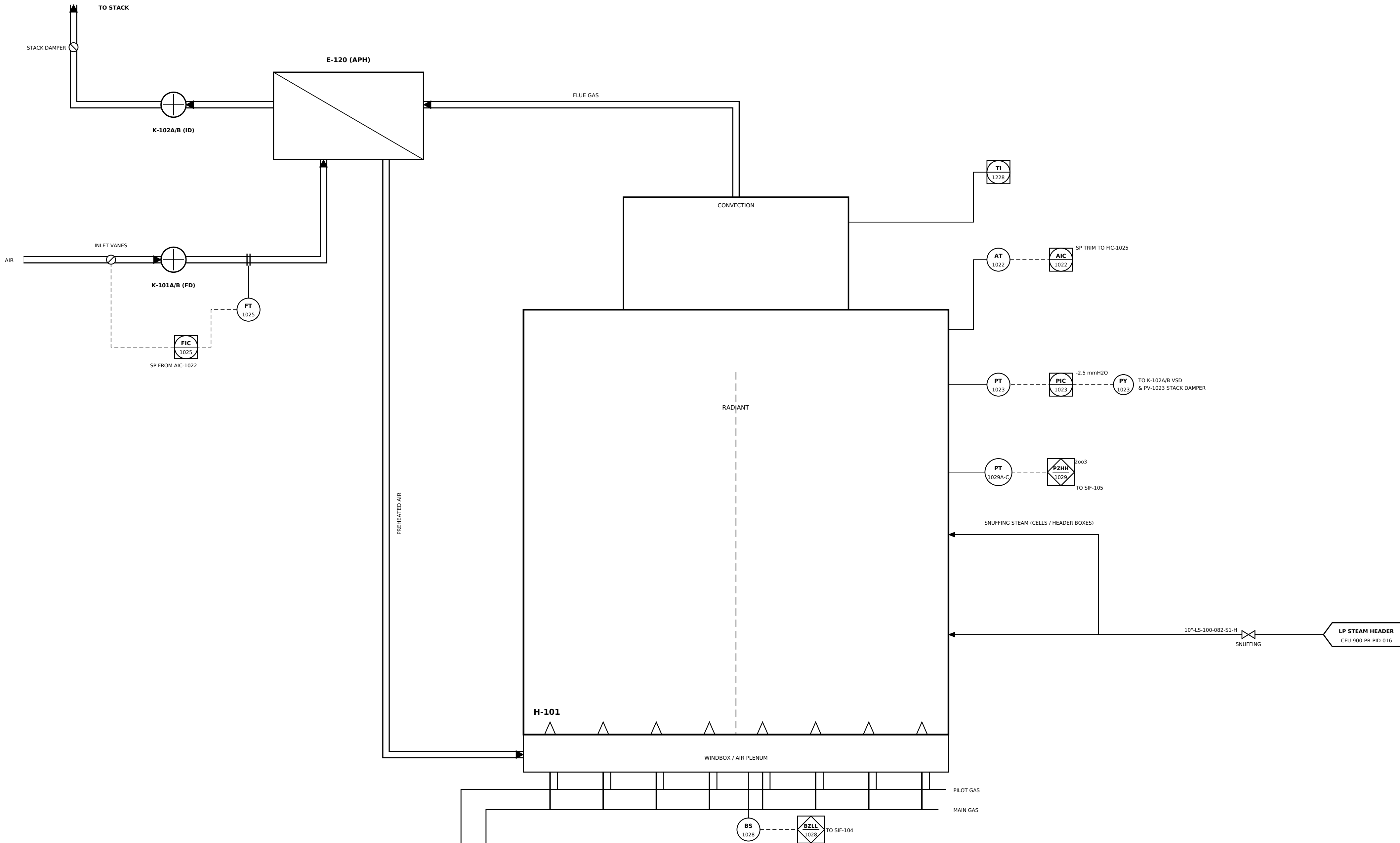
| REV                                        | DATE       | DESCRIPTION              |
|--------------------------------------------|------------|--------------------------|
| A                                          | 2026-10-02 | Issued for review (FEED) |
| GENERIC REFINERY - GULF COAST              |            |                          |
| 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT |            |                          |
| PIPING & INSTRUMENT DIAGRAM                |            |                          |
| ATMOSPHERIC HEATER H-101 - PROCESS COILS   |            |                          |
| DISCIPLINE                                 | STAGE      | SCALE                    |
| PROCESS                                    | FEED       | NTS                      |
|                                            |            | SHEET                    |
|                                            |            | 5 OF 17                  |
| DRAWN                                      | CHECKED    | APPROVED                 |
| CLAUDE CODE                                | -          | -                        |
|                                            |            | DATE                     |
|                                            |            | 2026-10-02               |
| DRAWING NUMBER                             |            | REV                      |
| CFU-100-PR-PID-004                         |            | A                        |

**H-101**  
Atmospheric crude charge heater  
60.7 MW abs. / 67.4 MW fired; 2-cell cabin, 8 passes, 144 rad.  
tubes 6" x 18.3 m  
DESIGN 38.5 barg / 545 C; CA 3 mm  
DUTY 60.68 MW  
MAT'L Tubes A335 P9 (9Cr-1Mo); convection P5 / CS studded

**E-120**  
H-101 cast-iron/glass-tube air preheater  
Flue gas 380 -> 160 C  
DESIGN None barg / 450 C  
MAT'L Cast iron / borosilicate

**K-101A/B**  
H-101 forced-draft fan  
96 t/h flue gas  
MOTOR 90 kW

**K-102A/B**  
H-101 induced-draft fan  
96 t/h flue gas  
MOTOR 200 kW

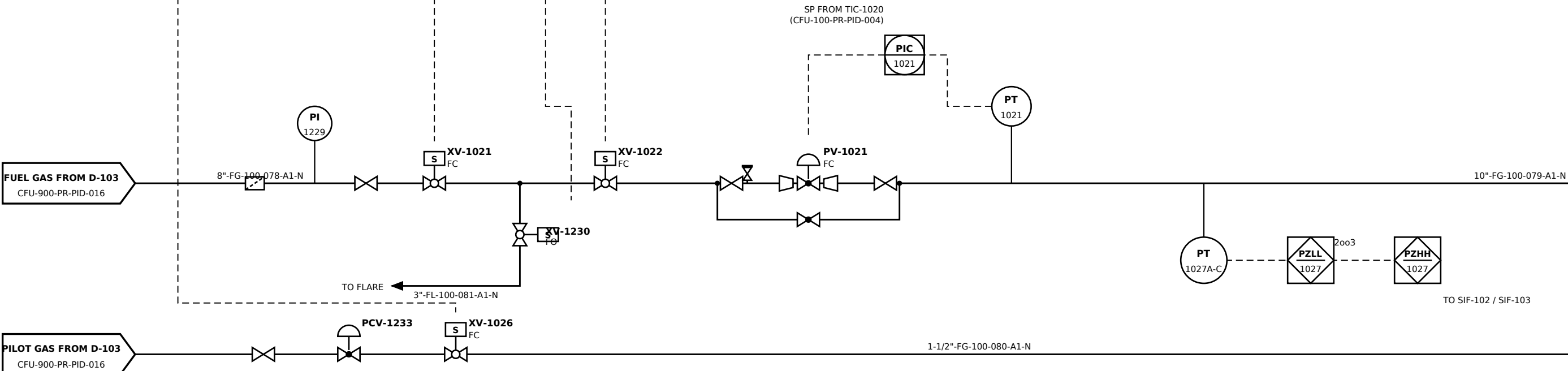


INPUTS: FZLL-1011..1018, TZHH-1020 (CFU-100-PR-PID-004), PZLL/PZHH-1027, BZLL-1028, PZHH-1029 (THIS SHEET)

**H-101 BMS / SIS (SIL-RATED PLC)**

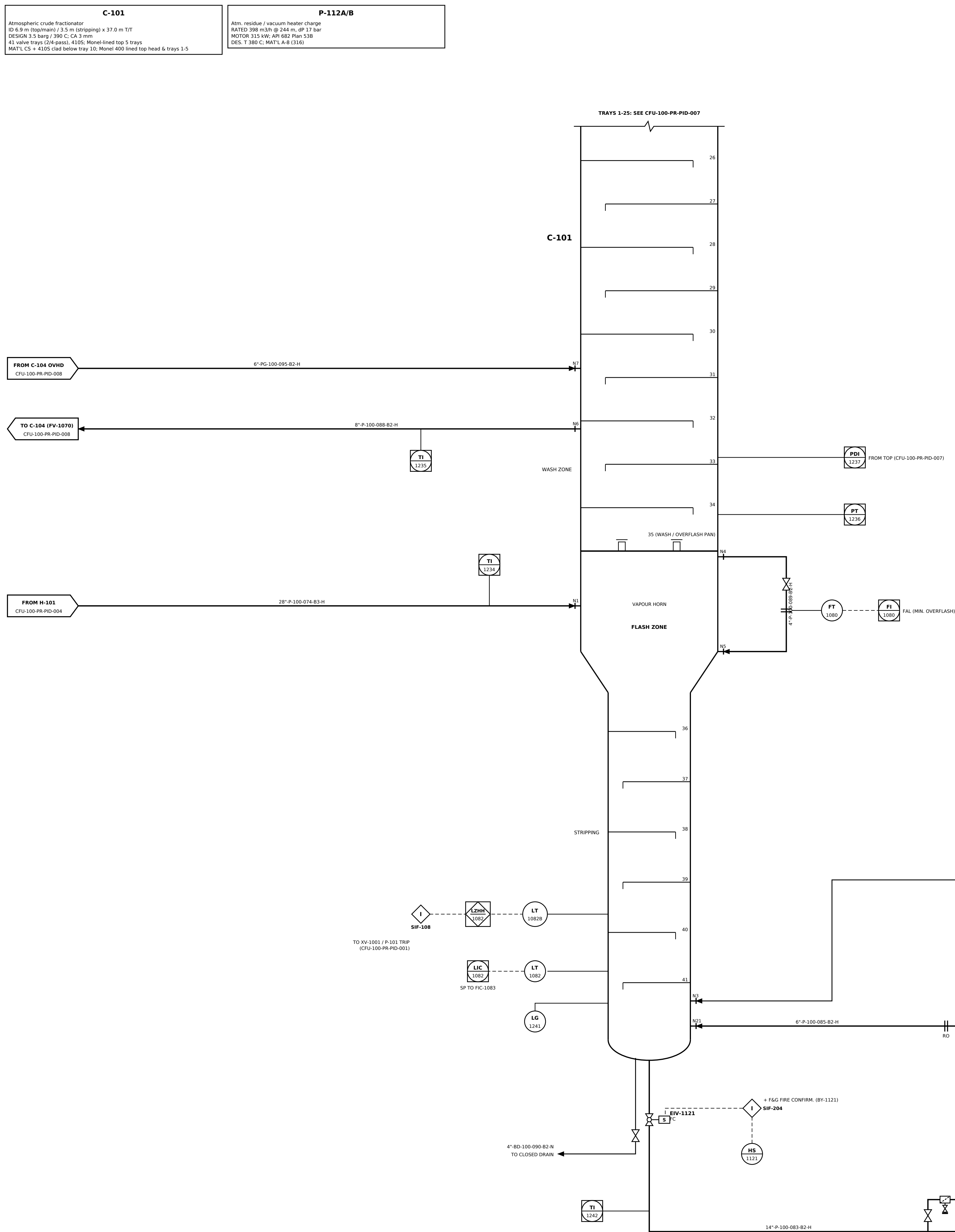
- SIF-101 (SIL 2)**  
H-101 low-low pass flow (any pass) -> trip fue
- SIF-102 (SIL 1)**  
H-101 low-low fuel gas pressure -> trip fuel
- SIF-103 (SIL 1)**  
H-101 high-high fuel gas pressure -> trip fuel
- SIF-104 (SIL 2)**  
H-101 loss of flame (all burners) -> trip fuel
- SIF-105 (SIL 1)**  
H-101 high-high arch pressure / loss of ID fan
- SIF-106 (SIL 1)**  
H-101 high-high COT -> trip fuel

TRIP: XV-1021/1022 + XV-1026 CLOSE, VENT XV OPEN



- NOTES**
- Burner management per API 556 / NFPA 85; purge, light-off and trip sequences in BMS (SIS logic).
  - Main fuel SSOVs XV-1021/1022 with vent XV (double block & bleed); all fail closed (vent fails open).
  - Fuel / air cross-limiting; AIC-1022 trims FIC-1025 set point; PIC-1021 low-select vs. minimum fire.
  - Snuffing steam manifold valves located >= 15 m from heater (API 560); one per cell + header boxes.
  - Ducts shown double-line; K-101A/B and K-102A/B are 2 x 100 % (one fan symbol shown per pair).
  - Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
  - All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
  - Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

| REV                                                                           | DATE       | DESCRIPTION              |               |
|-------------------------------------------------------------------------------|------------|--------------------------|---------------|
| A                                                                             | 2026-10-02 | Issued for review (FEED) |               |
| GENERIC REFINERY - GULF COAST<br>100 KBPSD CRUDE & VACUUM DISTILLATION UNIT   |            |                          |               |
| PIPING & INSTRUMENT DIAGRAM<br>ATMOSPHERIC HEATER H-101 - FIRING & AIR SYSTEM |            |                          |               |
| DISCIPLINE<br>PROCESS                                                         |            | STAGE<br>FEED            | SCALE<br>NTS  |
|                                                                               |            | SHEET<br>6 OF 17         |               |
| DRAWN<br>CLAUDE CODE                                                          |            | CHECKED<br>-             | APPROVED<br>- |
|                                                                               |            | DATE<br>2026-10-02       |               |
| DRAWING NUMBER<br>CFU-100-PR-PID-005                                          |            |                          | REV<br>A      |



- ## NOTES
- 1. Trays 1-25 and overhead on CFU-100-PR-PID-007. Striping section trays 36-41 (6 trays, 4.1 m ID).
  - 2. Overflash measured on external loop FI-1080 (minimum overflash alarm, APC controlled).
  - 3. EIV-1121 remotely operated isolation valve (ROSOV), fire-safe, actuated from CCR / SIF (SF-204).
  - 4. P-112 minimum flow returns to C-101 pump via RO.
  - 5. Line numbers: <num>-<fluid>-<area>-<seg>-<class>-<insulation>-per CONVENTIONS; see line list CFU-000-PI-LP-001.
  - 6. All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
  - 7. Instrument tags per ISA-S.1; include CFU-000-CIX-IDX-001. SIF functions per SRS [SIF tags shown in interlocks].

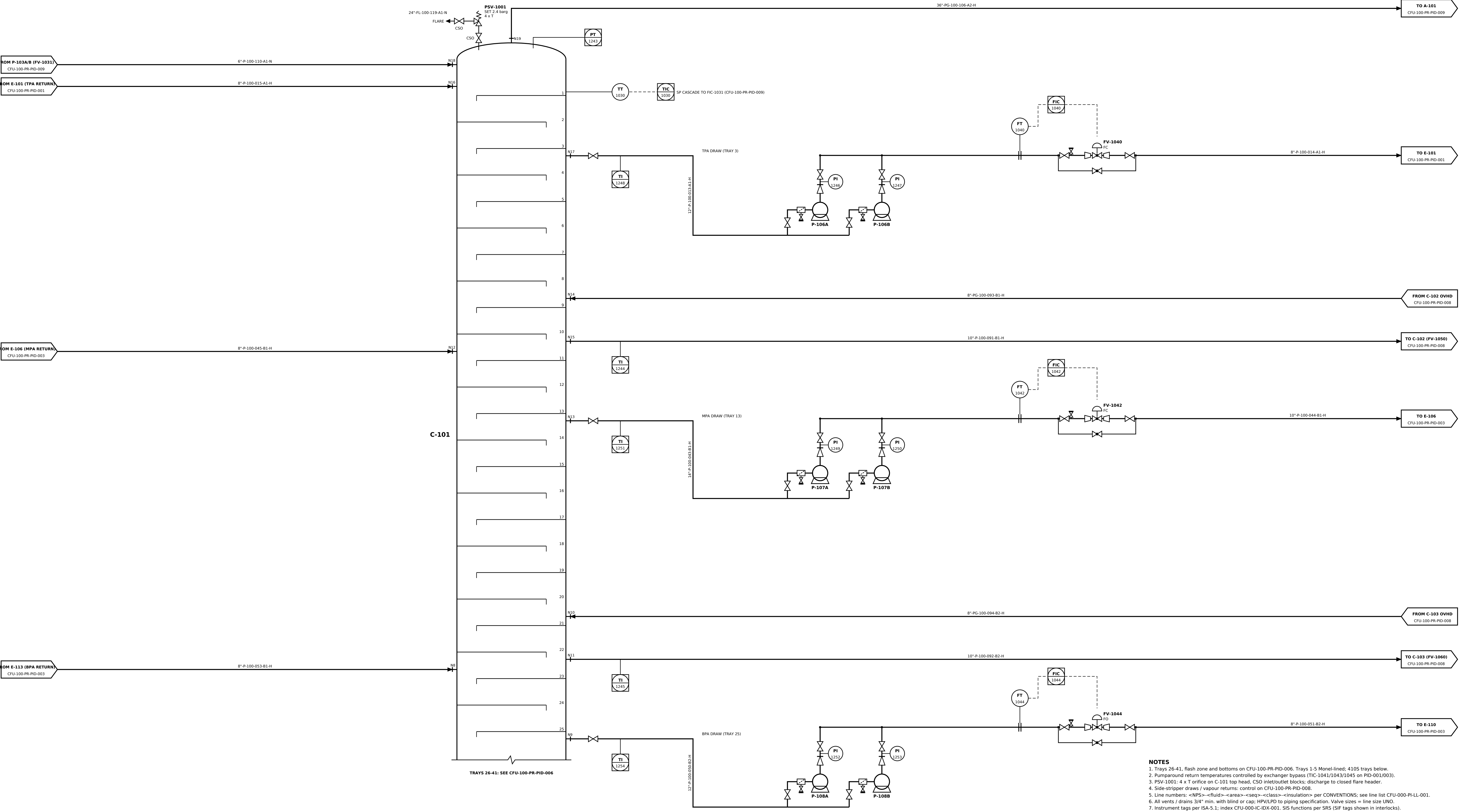
|                                                                                           |            |                          |            |          |
|-------------------------------------------------------------------------------------------|------------|--------------------------|------------|----------|
| REV                                                                                       | DATE       | DESCRIPTION              |            |          |
| A                                                                                         | 2026-10-02 | Issued for review (FEED) |            |          |
| GENERIC REFINERY - GULF COAST<br><b>100 KBPSD CRUDE &amp; VACUUM DISTILLATION UNIT</b>    |            |                          |            |          |
| <b>PIPING &amp; INSTRUMENT DIAGRAM</b><br>ATMOSPHERIC COLUMN C-101 - FLASH ZONE & BOTTOMS |            |                          |            |          |
| DISCIPLINE                                                                                | STAGE      | FEED                     | SCALE      | SHEET    |
| PROCESS                                                                                   |            |                          | NTS        | 7 OF 17  |
| DRAWN                                                                                     | CHECKED    | APPROVED                 | DATE       |          |
| CLAUDE CODE                                                                               | -          | -                        | 2026-10-02 |          |
| DRAWING NUMBER                                                                            |            |                          |            | REV      |
| <b>CFU-100-PR-PID-006</b>                                                                 |            |                          |            | <b>A</b> |

**C-101**  
Atmospheric crude fractionator  
ID 6.9 m (top/main) / 3.5 m (stripping) x 37.0 m T/T  
DESIGN 3.5 barg / 300 C; CA 3 mm  
41 valve trays (2/4-pass), 4105; Monel-lined top 5 trays  
MAT'L CS + 4105 clad below tray 10; Monel 400 lined top head & trays  
1-5

**P-106A/B**  
TPA pumparound  
RATED 293 m<sup>3</sup>/h @ 130 m, dP 8 bar  
MOTOR 1104 kW, API 682 Plus 1152  
DES. T 185 C; MAT'L S-6 (CS/12Cr)

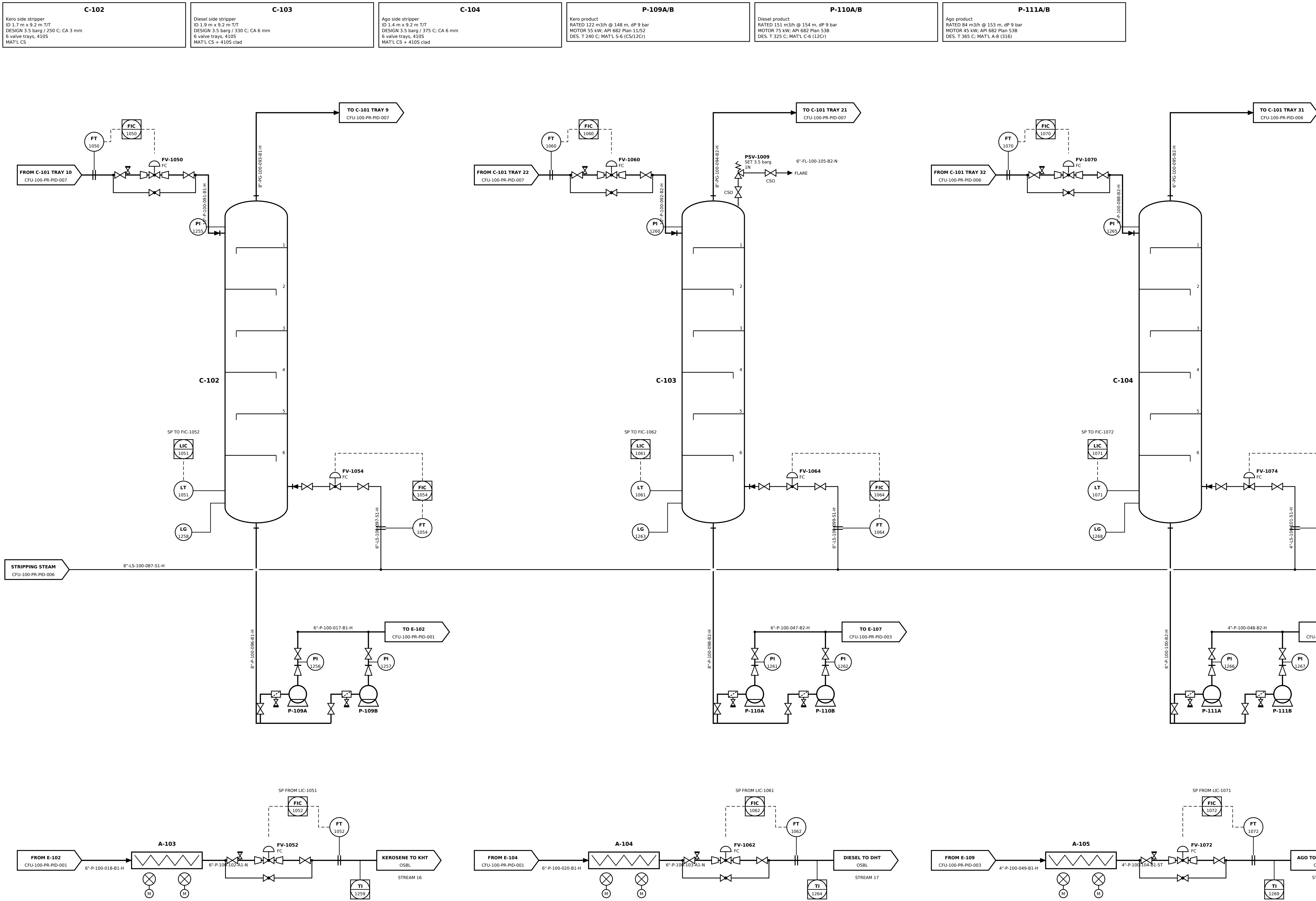
**P-107A/B**  
MPA pumparound  
RATED 387 m<sup>3</sup>/h @ 138 m, dP 8 bar  
MOTOR 1364 kW, API 682 Plus 1152  
DES. T 270 C; MAT'L C-6 (12Cr)

**P-108A/B**  
BPA pumparound  
RATED 318 m<sup>3</sup>/h @ 142 m, dP 8 bar  
MOTOR 1324 kW, API 682 Plus 538  
DES. T 345 C; MAT'L A-8 (316)



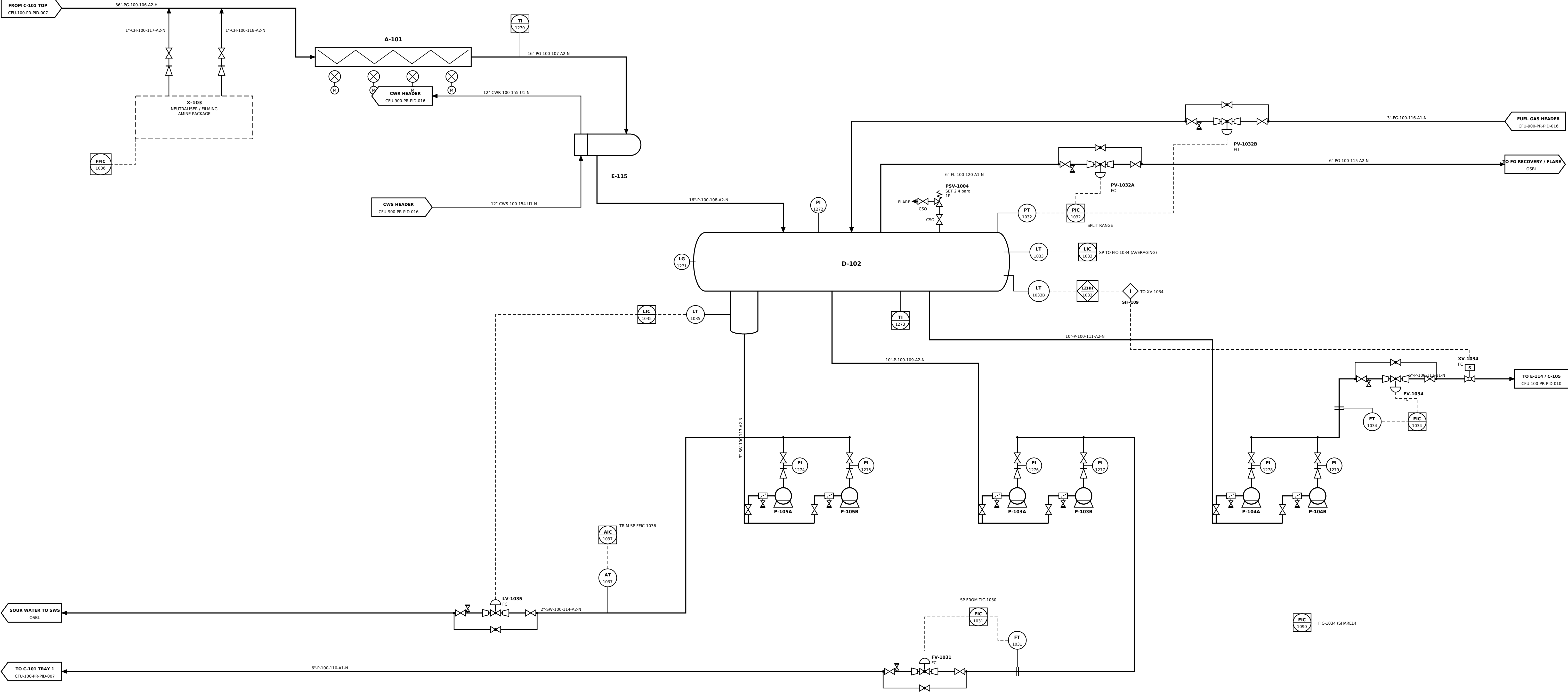
- NOTES**
- Trays 26-41, flash zone and bottoms on CFU-100-PR-PID-006. Trays 1-5 Monel-lined; 4105 trays below.
  - Pumparound return temperatures controlled by exchanger bypass (TIC-1041/1043/1045 on PID-001/003).
  - PSV-1001: 4 x T orifice on C-101 top head. CSO inlet/outlet blocks: discharge to closed flare header.
  - Side-stripper draws / vapour returns: control on CFU-100-PR-PID-008.
  - Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
  - All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
  - Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

| REV                                                    | DATE       | DESCRIPTION              |            |
|--------------------------------------------------------|------------|--------------------------|------------|
| A                                                      | 2026-10-02 | Issued for review (FEED) |            |
| GENERIC REFINERY - GULF COAST                          |            |                          |            |
| 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT             |            |                          |            |
| PIPING & INSTRUMENT DIAGRAM                            |            |                          |            |
| ATMOSPHERIC COLUMN C-101 - UPPER SECTION & PUMPAROUNDS |            |                          |            |
| DISCIPLINE                                             | STAGE      | SCALE                    | SHEET      |
| PROCESS                                                | FEED       | NTS                      | 8 OF 17    |
| DRAWN                                                  | CHECKED    | APPROVED                 | DATE       |
| CLAUDE CODE                                            | -          | -                        | 2026-10-02 |
| DRAWING NUMBER                                         |            |                          | REV        |
| CFU-100-PR-PID-007                                     |            |                          | A          |



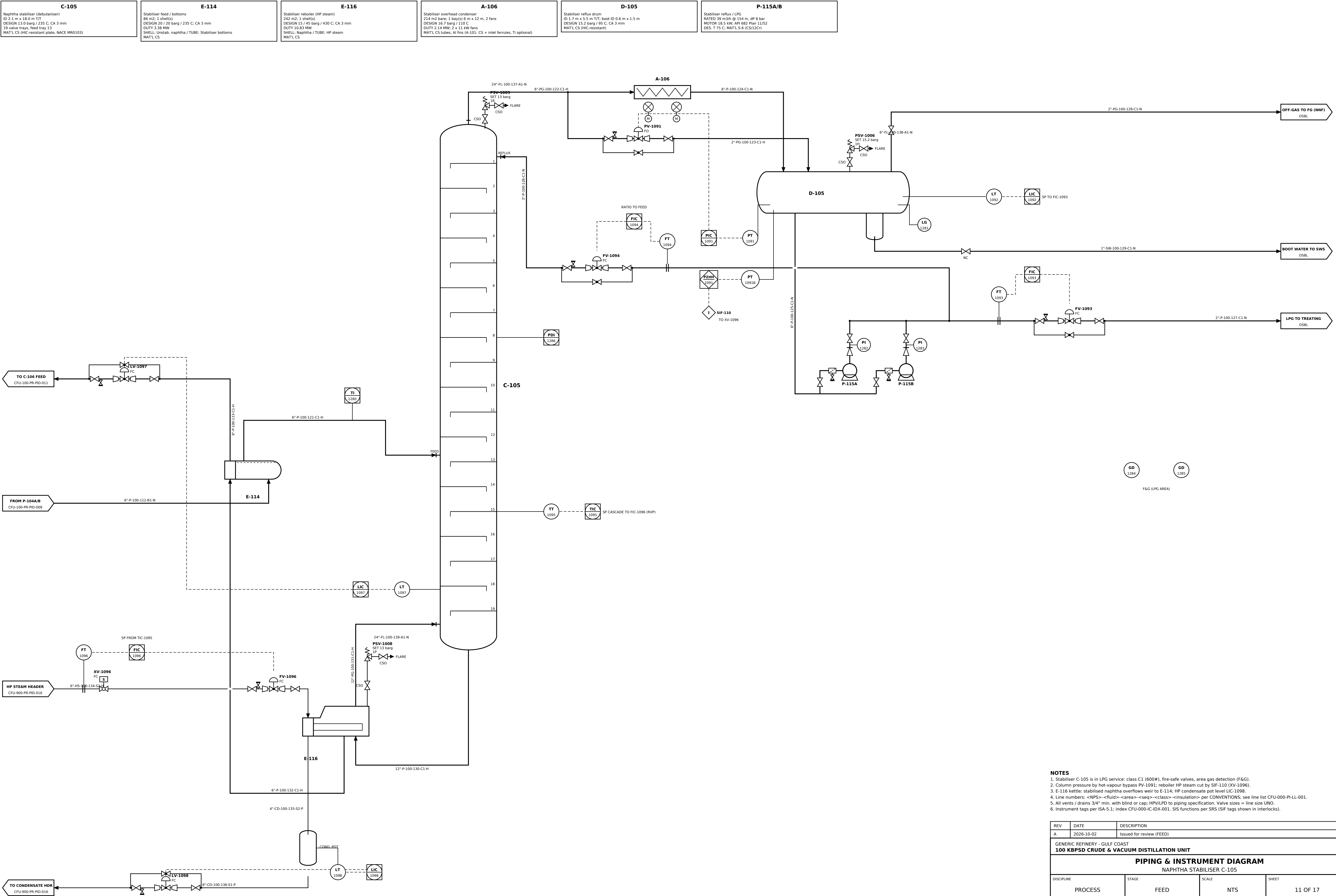
| REV                                                                           |         | DATE       | DESCRIPTION              |
|-------------------------------------------------------------------------------|---------|------------|--------------------------|
| A                                                                             |         | 2026-10-02 | Issued for review (FEED) |
| GENERIC REFINERY - GULF COAST<br>100 KBPSD CRUDE & VACUUM DISTILLATION UNIT   |         |            |                          |
| PIPING & INSTRUMENT DIAGRAM<br>SIDE STRIPPERS C-102/103/104 & PRODUCT RUNDOWN |         |            |                          |
| DISCIPLINE                                                                    | STAGE   | SCALE      | SHEET                    |
| PROCESS                                                                       | FEED    | NTS        | 9 OF 17                  |
| DRAWN                                                                         | CHECKED | APPROVED   | DATE                     |
| CLAUDE CODE                                                                   | -       | -          | 2026-10-02               |
| DRAWING NUMBER                                                                |         |            | REV                      |
| CFU-100-PR-PID-008                                                            |         |            | A                        |

|                                                                                                                                                                                                                       |                                                                                                                                                                                                             |                                                                                                                                                                              |                                                                                                                                            |                                                                                                                                                       |                                                                                                                                                   |                                                                                                                                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| <b>A-101</b><br>Atm. overhead condenser<br>1667 m2 bare; 3 bay(s) 6 m x 12 m, 6 fans<br>DESIGN 9.7 barg / 165 C<br>DUTY 33.75 MW; 6 x 55 kW fans<br>MAT'L CS tubes, Al fins (A-101: CS + inlet ferrules, Ti optional) | <b>E-115</b><br>Atm. overhead trim condenser<br>740 m2; 2 shell(s)<br>DESIGN 3.5 / 7 barg / 90 C; CA 3 mm<br>DUTY 5.96 MW<br>SHELL: OH vapour/condensate / TUBE: Cooling water<br>MAT'L CS shell / Ti tubes | <b>D-102</b><br>Atm. overhead reflux drum (3-phase)<br>ID 3.4 m x 12.0 m T/T; boot ID 1.5 m x 3.0 m<br>DESIGN 3.5 barg / 160 C; CA 6 mm<br>MAT'L CS (HIC-resistant), 6 mm CA | <b>P-103A/B</b><br>Atm. reflux<br>RATED 159 m3/h @ 102 m, dP 7 bar<br>MOTOR: 55 kW; API 682 Plan 11/52<br>DES. T 75 C; MAT'L S-6 (CS/L2Cr) | <b>P-104A/B</b><br>Unstabilised naphtha<br>RATED 180 m3/h @ 232 m, dP 16 bar<br>MOTOR: 160 kW; API 682 Plan 11/52<br>DES. T 75 C; MAT'L S-6 (CS/L2Cr) | <b>P-105A/B</b><br>Overhead sour water<br>RATED 14 m3/h @ 51 m, dP 2 bar<br>MOTOR: 5.5 kW; API 682 Plan 11/52<br>DES. T 75 C; MAT'L S-6 (CS/L2Cr) | <b>X-103</b><br>Neutraliser / filming amine package (atm OH)<br>Tank 2 m3 + 2 x 100 % metering pumps<br>MOTOR: 0.75 kW<br>MAT'L SS316 |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|



- NOTES**
- Neutraliser and filming amine injected upstream of A-101; pH control AIC-1037 trims FFIC-1036.
  - PIC-1032 split range: 0-50 % PV-1032B (FG make-up) closing, 50-100 % PV-1032A (off-gas) opening.
  - SIF-109 initiator / final element per SRS: LT-1033B high-high closes XV-1034 (see data issue log).
  - Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
  - All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
  - Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

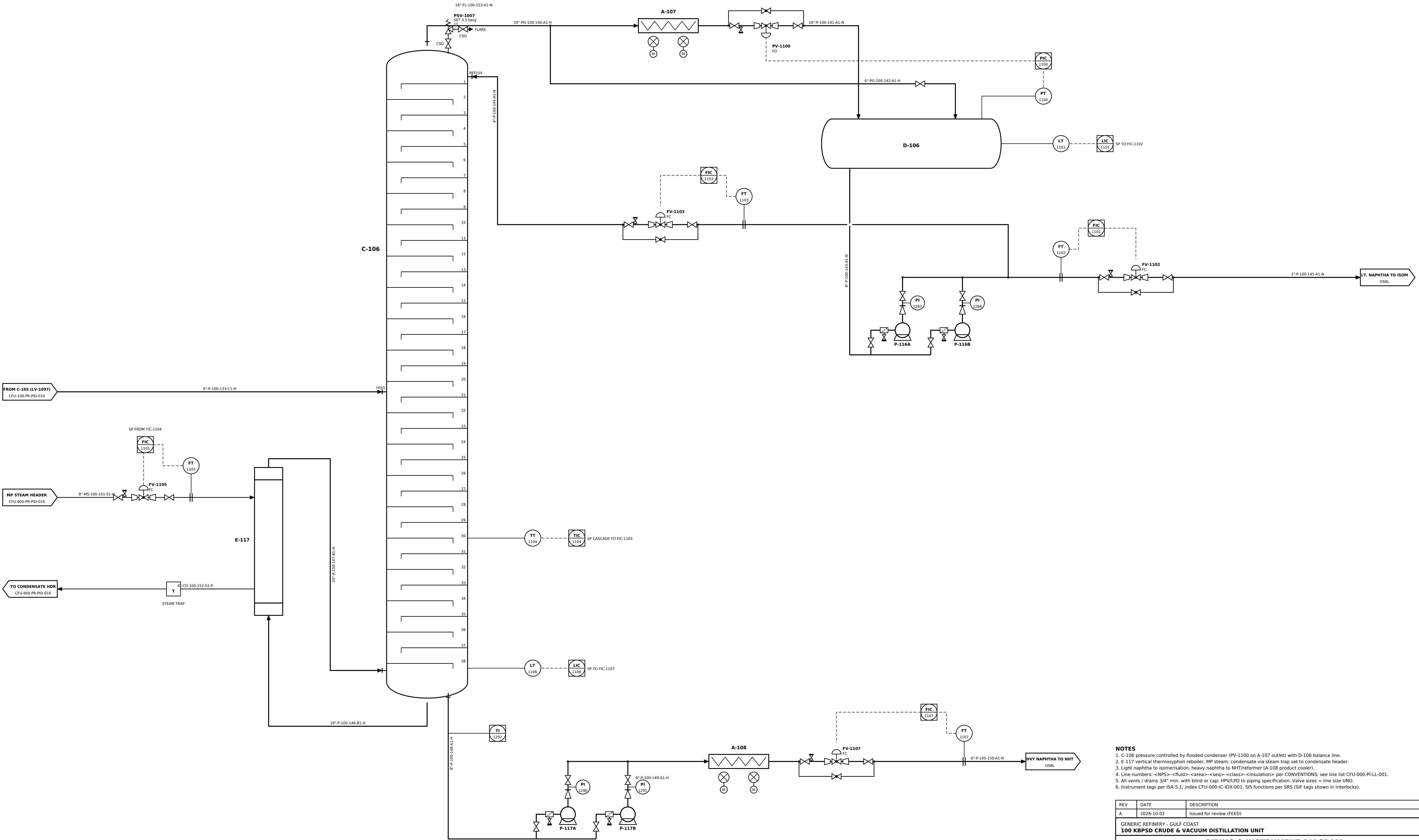
| REV                                        | DATE       | DESCRIPTION              |            |
|--------------------------------------------|------------|--------------------------|------------|
| A                                          | 2026-10-02 | Issued for review (FEED) |            |
| GENERIC REFINERY - GULF COAST              |            |                          |            |
| 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT |            |                          |            |
| PIPING & INSTRUMENT DIAGRAM                |            |                          |            |
| C-101 OVERHEAD SYSTEM & REFLUX DRUM D-102  |            |                          |            |
| DISCIPLINE                                 | STAGE      | SCALE                    | SHEET      |
| PROCESS                                    | FEED       | NTS                      | 10 OF 17   |
| DRAWN                                      | CHECKED    | APPROVED                 | DATE       |
| CLAUDE CODE                                | -          | -                        | 2026-10-02 |
| DRAWING NUMBER                             |            |                          | REV        |
| CFU-100-PR-PID-009                         |            |                          | A          |



- NOTES**
- Stabiliser C-105 is in LPG service; class C1 (600#), fire-safe valves, area gas detection (F&G).
  - Column pressure by hot-vapour bypass FV-1091; reboiler HP steam cut by SIF-110 (XV-1096).
  - E-116 kettle: stabilised naphtha overflows weir to E-114; HP condensate pot level LIC-1098.
  - Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
  - All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
  - Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

| REV                                        | DATE       | DESCRIPTION              |            |
|--------------------------------------------|------------|--------------------------|------------|
| A                                          | 2026-10-02 | Issued for review (FEED) |            |
| GENERIC REFINERY - GULF COAST              |            |                          |            |
| 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT |            |                          |            |
| PIPING & INSTRUMENT DIAGRAM                |            |                          |            |
| NAPHTHA STABILISER C-105                   |            |                          |            |
| DISCIPLINE                                 | STAGE      | SCALE                    | SHEET      |
| PROCESS                                    | FEED       | NTS                      | 11 OF 17   |
| DRAWN                                      | CHECKED    | APPROVED                 | DATE       |
| CLAUDE CODE                                | -          | -                        | 2026-10-02 |
| DRAWING NUMBER                             |            |                          | REV        |
| CFU-100-PR-PID-010                         |            |                          | A          |

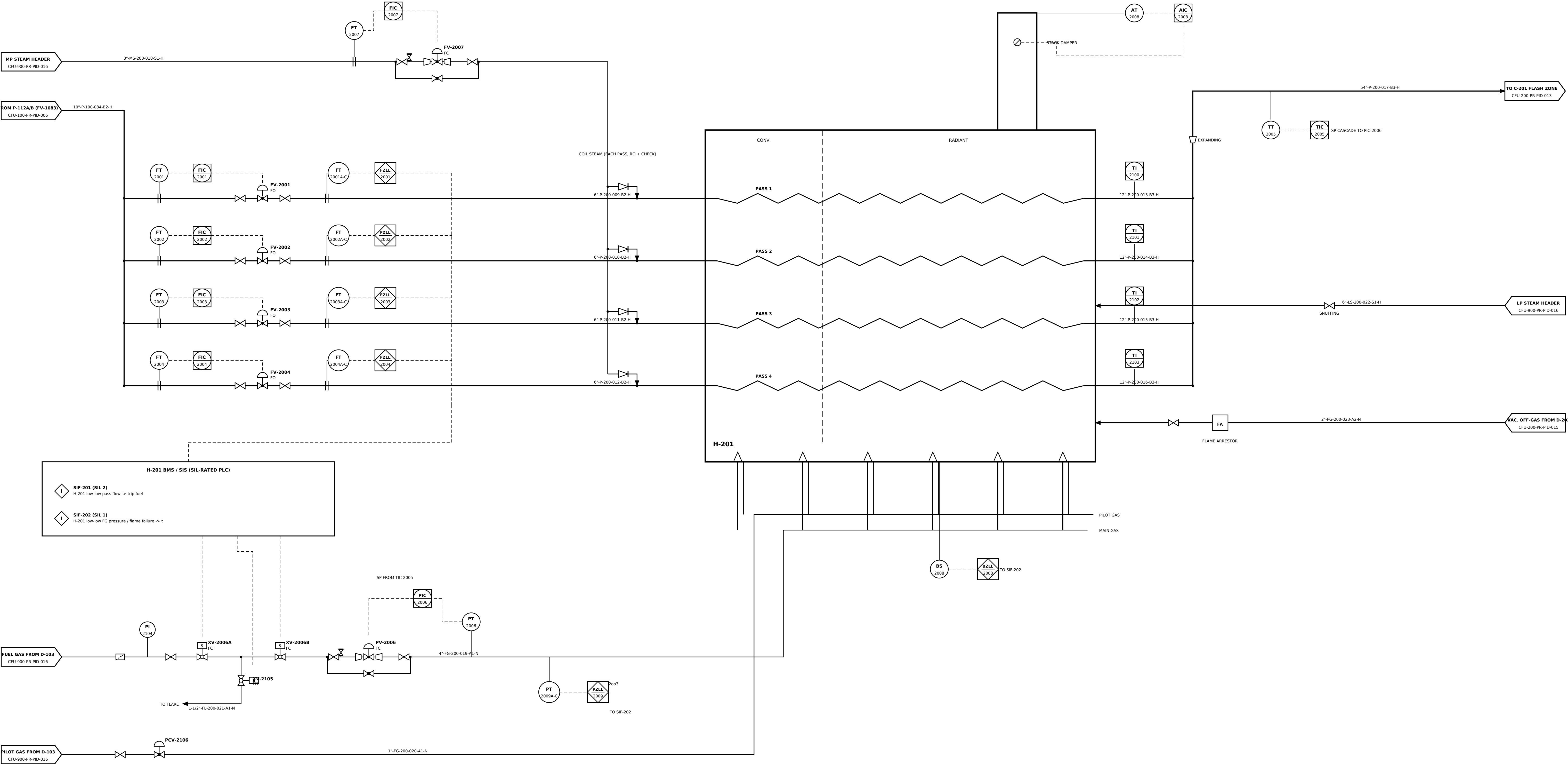
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|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>C-106</b><br>Naphtha splitter<br>ID 3.1 m x 29.5 m TTT<br>DESIGN 3.5 barg / 185 C; CA 3 mm<br>38 valve trays, feed tray 21<br>MAT'L CS | <b>E-117</b><br>Splitter reboiler (MP steam)<br>225 m2; 1 shell(s)<br>DESIGN 5 / 13 barg / 280 C; CA 3 mm<br>DUTY 6.83 MW<br>SHELL: Naphtha / TUBE: MP steam<br>MAT'L CS | <b>A-107</b><br>Splitter overhead condenser<br>BB1 m2 bare; 2 bay(s) 6 m x 12 m, 4 fans<br>DESIGN 9.7 barg / 135 C<br>DUTY 9.63 MW; 4 x 30 kW fans<br>MAT'L CS tubes, Al fins (A-101; CS + inlet ferrules, Ti optional) | <b>D-106</b><br>Splitter reflux drum<br>ID 2.7 m x 8.5 m TTT<br>DESIGN 3.5 barg / 120 C; CA 3 mm<br>MAT'L CS | <b>P-116A/B</b><br>Splitter reflux / LN<br>RATED 146 m3/h @ 111 m, dP 7 bar<br>MOTOR 55 kW, API 682 Flan 11/52<br>DES. T 80 C; MAT'L S-6 (CS/12Cr) | <b>P-117A/B</b><br>Heavy naphtha product<br>RATED 142 m3/h @ 98 m, dP 6 bar<br>MOTOR 45 kW, API 682 Flan 11/52<br>DES. T 185 C; MAT'L S-6 (CS/12Cr) | <b>A-108</b><br>Heavy naphtha product cooler<br>401 m2 bare; 1 bay(s) 6 m x 12 m, 2 fans<br>DESIGN 9.7 barg / 185 C<br>DUTY 5.81 MW; 2 x 30 kW fans<br>MAT'L CS tubes, Al fins (A-101; CS + inlet ferrules, Ti optional) |
|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



- NOTES**
- C-106 pressure controlled by flooded condenser (PV-1100 on A-107 outlet) with D-106 balance line.
  - E-117 vertical thermosyphon reboiler, MP steam; condensate via steam trap set to condensate header.
  - Light naphtha to isomerisation, heavy naphtha to NHT/reformer (A-108 product cooler).
  - Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
  - All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
  - Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

| REV                                                                         | DATE       | DESCRIPTION              |            |
|-----------------------------------------------------------------------------|------------|--------------------------|------------|
| A                                                                           | 2026-10-02 | Issued for review (FEED) |            |
| GENERIC REFINERY - GULF COAST<br>100 KBPSD CRUDE & VACUUM DISTILLATION UNIT |            |                          |            |
| PIPING & INSTRUMENT DIAGRAM<br>NAPHTHA SPLITTER C-106                       |            |                          |            |
| DISCIPLINE                                                                  |            | STAGE                    | SCALE      |
| PROCESS                                                                     |            | FEED                     | NTS        |
|                                                                             |            |                          | 12 OF 17   |
| DRAWN                                                                       |            | CHECKED                  | DATE       |
| CLAUDE CODE                                                                 |            | -                        | 2026-10-02 |
| DRAWING NUMBER                                                              |            |                          | REV        |
| CFU-100-PR-PID-011                                                          |            |                          | A          |

**H-201**  
Vacuum heater  
14.7 MW abs. / 17.5 MW fired; 1-cell cabin, 4 passes, 44 rad. tubes 6" x 18.3 m  
DESIGN 19.8 barg / 580 C; CA 3 mm  
DUTY 14.69 MW  
MAT'L Tubes A335 P9 (9Cr-1Mo); convection P5 / CS studded



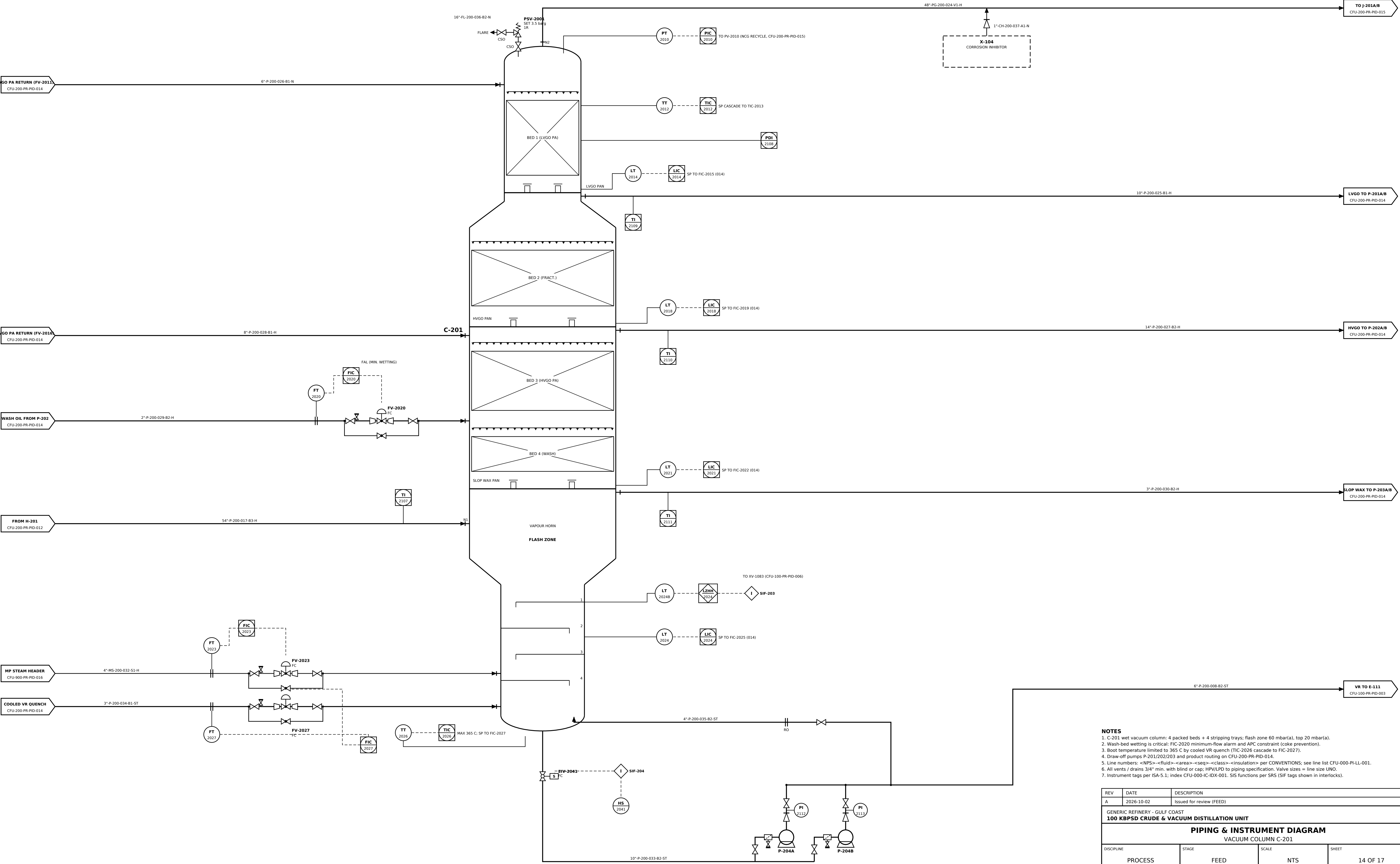
- NOTES**
- Passes shown schematically (convection + radiant in series); 4 identical passes, coil steam per pass.
  - Pass flow SIS transmitters FT-200xA/B/C (2003) independent of BPCS FT-200x (SIF-201).
  - Vacuum off-gas burned in dedicated H-201 burners via flame arrestor (from D-202 / PV-2031).
  - Natural draft heater: O2 trim AIC-2008 on stack damper.
  - Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
  - All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
  - Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

| REV                                        | DATE       | DESCRIPTION              |            |
|--------------------------------------------|------------|--------------------------|------------|
| A                                          | 2026-10-02 | Issued for review (FEED) |            |
| GENERIC REFINERY - GULF COAST              |            |                          |            |
| 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT |            |                          |            |
| PIPING & INSTRUMENT DIAGRAM                |            |                          |            |
| VACUUM HEATER H-201                        |            |                          |            |
| DISCIPLINE                                 |            | STAGE                    | SCALE      |
| PROCESS                                    |            | FEED                     | NTS        |
|                                            |            | SHEET                    |            |
|                                            |            | 13 OF 17                 |            |
| DRAWN                                      | CHECKED    | APPROVED                 | DATE       |
| CLAUDE CODE                                | -          | -                        | 2026-10-02 |
| DRAWING NUMBER                             |            |                          | REV        |
| CFU-200-PR-PID-012                         |            |                          | A          |

**C-201**  
Vacuum column (wet, packed)  
ID 4.4 m (top) / 6.6 m (main) / 3.0 m (boot) x 36.0 m T/T  
DESIGN PV: 3.5 barg / 435 C; CA 6 mm  
4 packed beds (structured 250V/grid), 4 stripping trays, vapour horn  
MATL: CS + 410S clad (317L clad flash/wash zone)

**P-204A/B**  
Vacuum residue (incl. quench)  
RATED 196 m<sup>3</sup>/h @ 187 m, dP 14 bar  
MOTOR: 132 kW; API 682 Plan 538  
DES. T 395 C; MATL: A-8 (316)

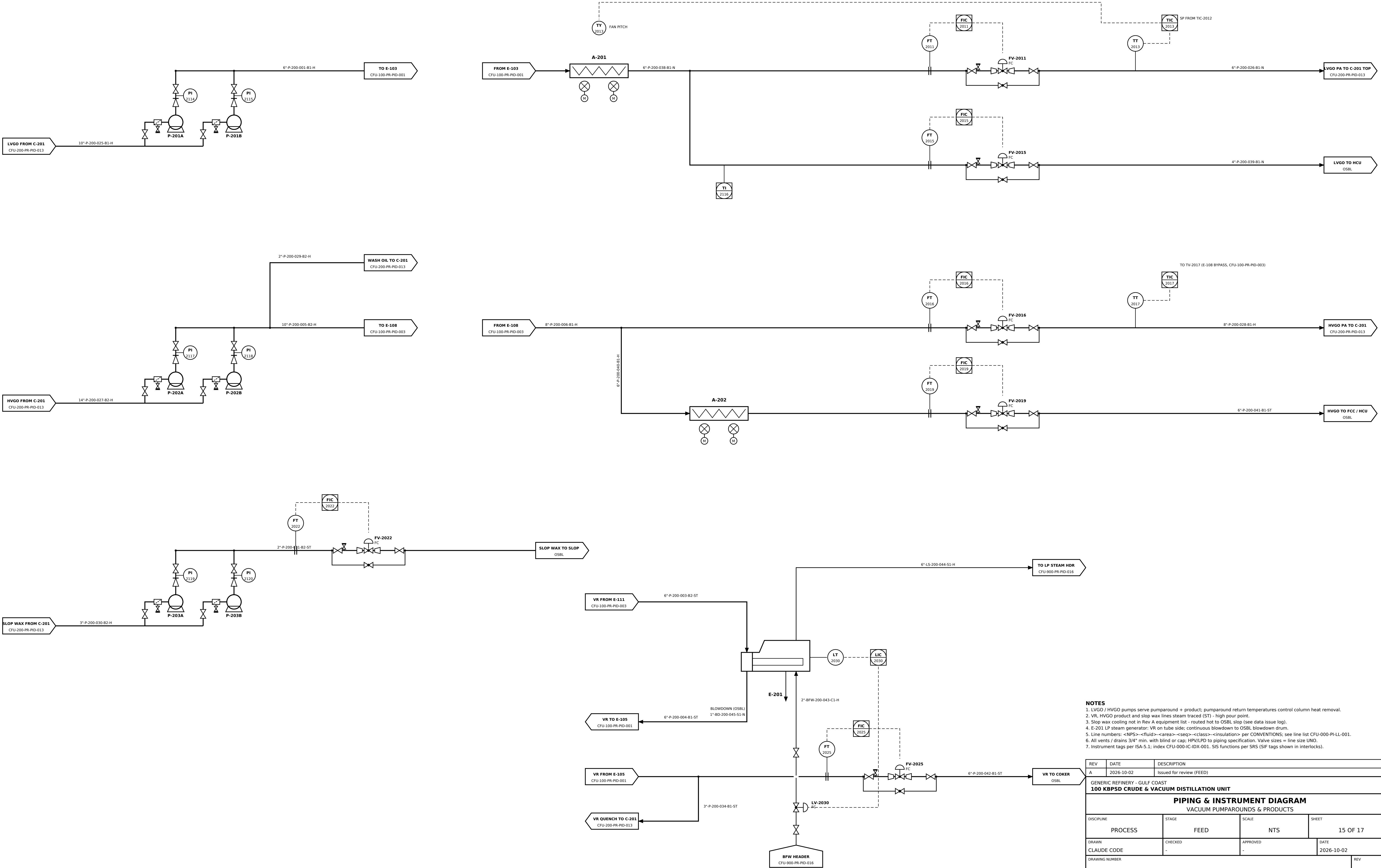
**X-104**  
Corrosion inhibitor package (VDU OH)  
Tank 2 m<sup>3</sup> + 2 x 100 % metering pumps  
MOTOR: 0.75 kW  
MATL: SS316

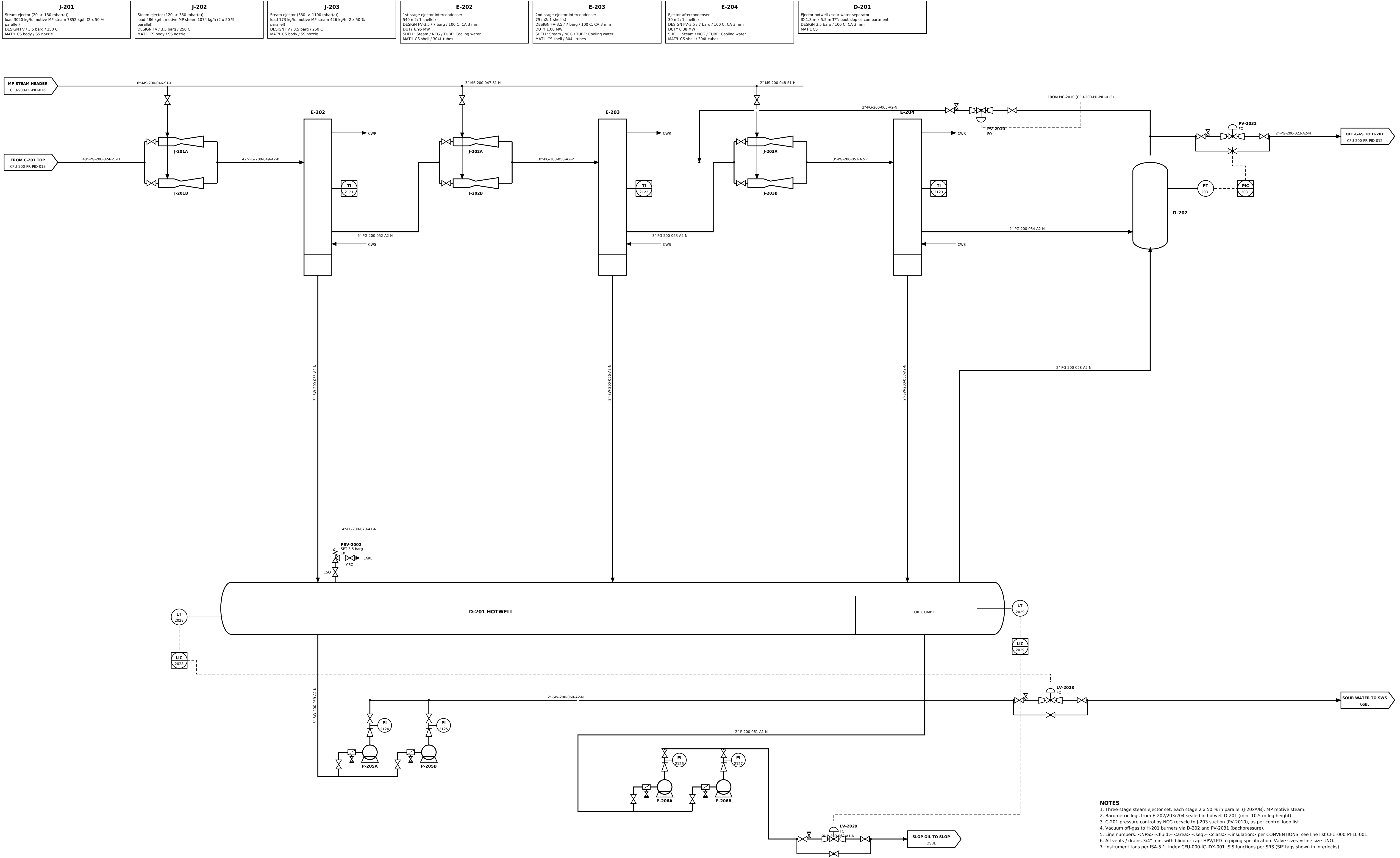


- NOTES**
- C-201 wet vacuum column: 4 packed beds + 4 stripping trays; flash zone 60 mbar(a), top 20 mbar(a).
  - Wash-bed wetting is critical: FIC-2020 minimum-flow alarm and APC constraint (coke prevention).
  - Boot temperature limited to 365 C by cooled VR quench (TIC-2026 cascade to FIC-2027).
  - Draw-off pumps P-201/202/203 and product routing on CFU-200-PR-PID-014.
  - Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
  - All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
  - Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

| REV                                        | DATE       | DESCRIPTION              |            |
|--------------------------------------------|------------|--------------------------|------------|
| A                                          | 2026-10-02 | Issued for review (FEED) |            |
| GENERIC REFINERY - GULF COAST              |            |                          |            |
| 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT |            |                          |            |
| PIPING & INSTRUMENT DIAGRAM                |            |                          |            |
| VACUUM COLUMN C-201                        |            |                          |            |
| DISCIPLINE                                 | STAGE      | SCALE                    | SHEET      |
| PROCESS                                    | FEED       | NTS                      | 14 OF 17   |
| DRAWN                                      | CHECKED    | APPROVED                 | DATE       |
| CLAUDE CODE                                | -          | -                        | 2026-10-02 |
| DRAWING NUMBER                             |            |                          | REV        |
| CFU-200-PR-PID-013                         |            |                          | A          |

|                                                                                                                                                           |                                                                                                                                                      |                                                                                                                                   |                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>P-201A/B</b><br>LVGO pumparound / product<br>RATED 198 m3/h @ 138 m, dP 10 bar<br>MOTOR 110 kW, API 682 Plan 1122<br>DES. T 255 C; MAT'L S-6 (CS/12Cr) | <b>P-202A/B</b><br>HVGO pumparound / product<br>RATED 391 m3/h @ 144 m, dP 10 bar<br>MOTOR 200 kW, API 682 Plan 538<br>DES. T 345 C; MAT'L A-8 (316) | <b>P-203A/B</b><br>Slop wax<br>RATED 12 m3/h @ 114 m, dP 8 bar<br>MOTOR 5.5 kW, API 682 Plan 538<br>DES. T 410 C; MAT'L A-8 (316) | <b>A-201</b><br>Lvgo cooler<br>397 m2 bare; 1 bay(s) 6 m x 12 m, 2 fans<br>DESIGN 9.7 barg / 125 C<br>DUTY 3.10 MW; 2 x 15 kW fans<br>MAT'L CS tubes, Al fins (A-101; CS + inlet ferrules, Ti optional) | <b>A-202</b><br>Hvgo Product cooler<br>361 m2 bare; 1 bay(s) 6 m x 12 m, 2 fans<br>DESIGN 9.7 barg / 125 C<br>DUTY 6.64 MW; 2 x 37 kW fans<br>SHELL: BFW / LP steam / TUBE: Vacuum residue<br>MAT'L CS tubes, Al fins (A-101; CS + inlet ferrules, Ti optional) | <b>E-201</b><br>VR / LP steam generator (kettle)<br>84 m2; 1 shell(s)<br>DESIGN 6 / 20 barg / 310 C; CA 3 mm<br>DUTY 1.78 MW<br>SHELL: BFW / LP steam / TUBE: Vacuum residue<br>MAT'L CS shell / Scr tubes |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|





COOLING WATER CONNECTIONS (HEADERS ON CFU-900-PR-PID-016):  
E-202: CWS 12" CWS-200-064-U1-N / CWR 12" CWR-200-065-U1-N  
E-203: CWS 6" CWS-200-066-U1-N / CWR 6" CWR-200-067-U1-N  
E-204: CWS 3" CWS-200-068-U1-N / CWR 3" CWR-200-069-U1-N

- NOTES**
- Three-stage steam ejector set, each stage 2 x 50 % in parallel (J-20xA/B); MP motive steam.
  - Barometric legs from E-202/203/204 sealed in hotwell D-201 (min. 10.5 m leg height).
  - C-201 pressure control by NCG recycle to J-203 suction (PV-2010), as per control loop list.
  - Vacuum off-gas to H-201 burners via D-202 and PV-2031 (backpressure).
  - Line numbers: <NPS>-<fluid>-<area>-<seq>-<class>-<insulation> per CONVENTIONS; see line list CFU-000-PI-LL-001.
  - All vents / drains 3/4" min. with blind or cap; HPV/LPD to piping specification. Valve sizes = line size UNO.
  - Instrument tags per ISA-5.1; index CFU-000-IC-IDX-001. SIS functions per SRS (SIF tags shown in interlocks).

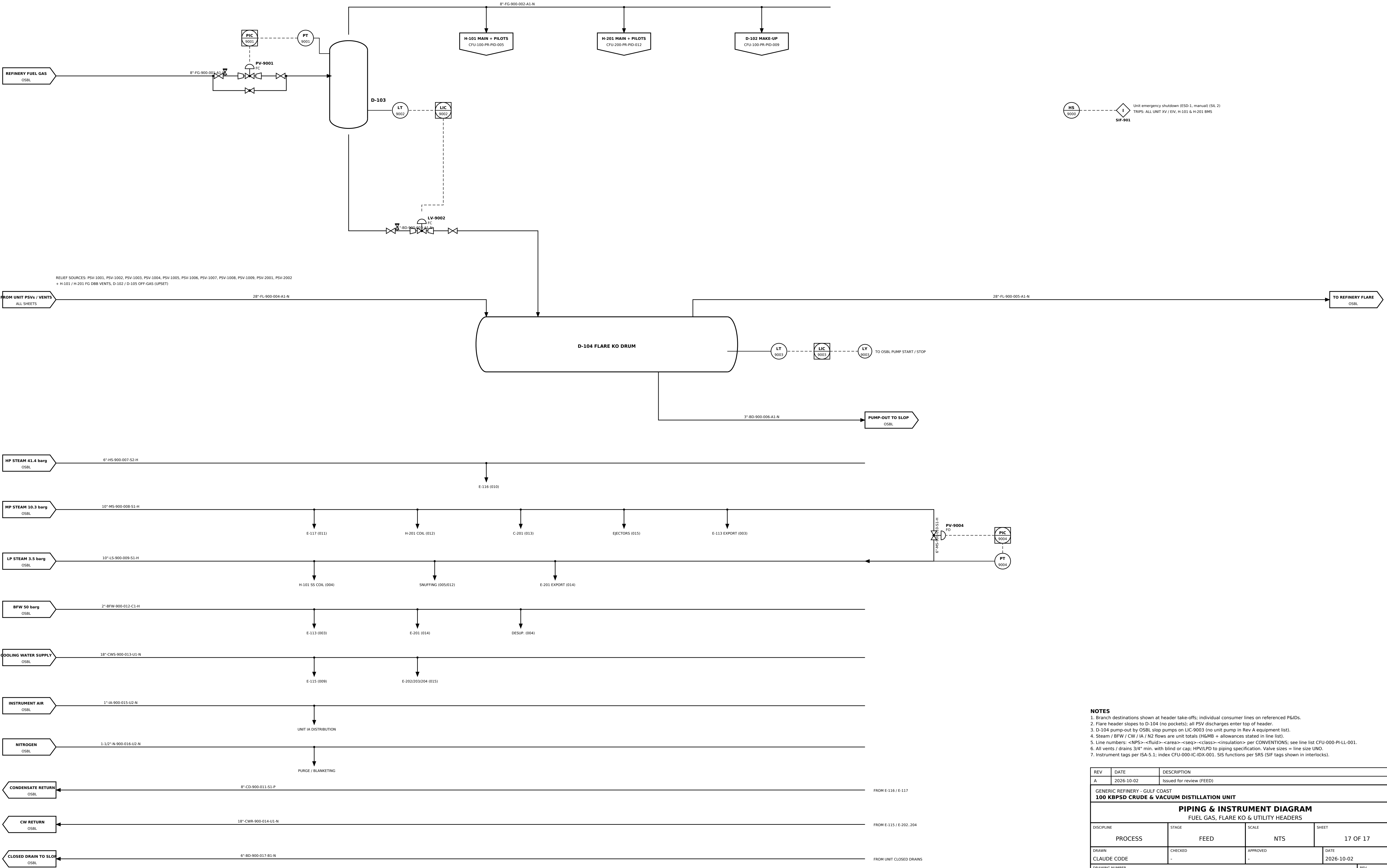
| REV                                        | DATE       | DESCRIPTION              |            |
|--------------------------------------------|------------|--------------------------|------------|
| A                                          | 2026-10-02 | Issued for review (FEED) |            |
| GENERIC REFINERY - GULF COAST              |            |                          |            |
| 100 KBPSD CRUDE & VACUUM DISTILLATION UNIT |            |                          |            |
| PIPING & INSTRUMENT DIAGRAM                |            |                          |            |
| VACUUM EJECTOR SYSTEM & HOTWELL D-201      |            |                          |            |
| DISCIPLINE                                 | STAGE      | SCALE                    | SHEET      |
| PROCESS                                    | FEED       | NTS                      | 16 OF 17   |
| DRAWN                                      | CHECKED    | APPROVED                 | DATE       |
| CLAUDE CODE                                | -          | -                        | 2026-10-02 |
| DRAWING NUMBER                             |            |                          | REV        |
| CFU-200-PR-PID-015                         |            |                          | A          |

D-103

Fuel gas knock-out drum  
ID 0.9 m x 2.5 m TTT  
DESIGN 7.0 barg / 90 C; CA 3 mm  
MAT'L CS

D-104

Flare knock-out drum (unit)  
ID 2.1 m x 6.5 m TTT  
DESIGN 3.5 barg / 280 C; CA 3 mm  
MAT'L CS



| REV                                                                         |         | DATE       | DESCRIPTION              |
|-----------------------------------------------------------------------------|---------|------------|--------------------------|
| A                                                                           |         | 2026-10-02 | Issued for review (FEED) |
| GENERIC REFINERY - GULF COAST<br>100 KBPSD CRUDE & VACUUM DISTILLATION UNIT |         |            |                          |
| PIPING & INSTRUMENT DIAGRAM<br>FUEL GAS, FLARE KO & UTILITY HEADERS         |         |            |                          |
| DISCIPLINE                                                                  | STAGE   | SCALE      | SHEET                    |
| PROCESS                                                                     | FEED    | NTS        | 17 OF 17                 |
| DRAWN                                                                       | CHECKED | APPROVED   | DATE                     |
| CLAUDE CODE                                                                 | -       | -          | 2026-10-02               |
| DRAWING NUMBER                                                              |         |            | REV                      |
| CFU-900-PR-PID-016                                                          |         |            | A                        |